

Recent advances in the rational designing of metalloporphyrinoid-based CO₂ reduction catalysts: From molecular structural tuning to the application in catalysis

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Dedicated to Prof. Tomás Torres on the occasion of his 70th birthday

ABSTRACT: Electro and photochemical CO₂ reduction (CO₂R) has emerged as a contemporary research field for tackling the fossil fuel problems and construction of a modern chemical industry. For efficient CO₂R, a catalyst is needed, and Earth-abundant Fe, Co, and Ni-based metalloporphyrinoids have appeared as promising molecular catalysts for CO₂R. Several metal complexes of porphyrins, phthalocyanines, corroles, and chlorins-type porphyrinoids have been designed and tested for this purpose. The unique advantages in the judicious tuning of the porphyrinoid structures and their susceptibility towards different functionalization reactions allow an impressive scope to explore and establish a direct structure-activity relationship of metalloporphyrinoids towards CO₂R. There have been rapid advances in this field in the last few years. For the homogeneous CO₂R, the electronic and steric aspects of the porphyrinoids are fine-controlled by installing favorable functional groups at the periphery of the porphyrinoids. On the other hand, in heterogeneous CO₂R, the design principle is guided by integrating metalloporphyrinoids in composite material after mixing with carbon supports, semiconductors, or adopting reticular chemistry. All the advancements further result in the testing of excellent metalloporphyrinoids in industrially feasible CO₂ electrolyzers or photoelectrochemical devices. This review focuses on the recent advances in designing principles of metalloporphyrinoids catalysts in homogeneous, heterogeneous, and CO₂R reactors.

KEYWORDS: metalloporphyrin, metallophthalocyanine, metallocorrole, metallochlorin, CO₂ reduction, electrolyzer.

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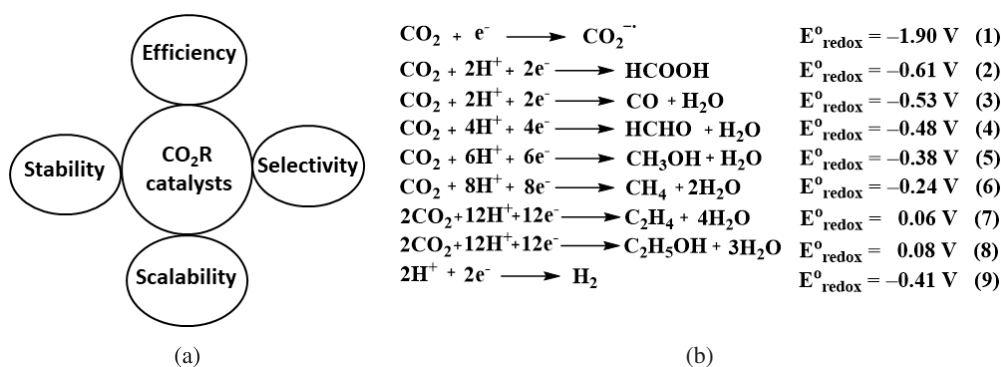


Fig. 1. (a) Key metrics of a CO₂R catalyst, (b) Thermodynamic formal potentials for various CO₂ reduction products. (pH 7 in aqueous solutions vs. NHE).

INTRODUCTION

Over the last few decades, research on electro, photo, or photo(electro)chemical CO₂ reduction (CO₂R) has witnessed substantial efforts towards generating fuels and value-added chemicals from CO₂ through ‘Power-to-X’ approaches [1]. CO₂R technologies are envisioned to circumvent the fossil fuel dependence for energy needs and establish a complementary solar fuels industry with a low carbon footprint economy [2, 3]. Two basic industrial frameworks have emerged: (1) either having a fully integrated electro/photoreactor device or (2) an electrolyzer coupled to external photovoltaics (PV) or wind turbines, hydroelectricity, etc. However, the commercialization of CO₂R devices demands concerted multidisciplinary efforts from basic material and theoretical research to key engineering development on the reactor design side [4, 5]. Presently, at the electrode and device level, CO₂ conversion efficiency, product selectivity, stability, and scalability are predominant scientific challenges that need to be optimally addressed (Fig. 1a) [6].

Moreover, while scaling up the devices, the metrics must be conserved, maintaining an acceptable cost and abundance of the components. The accomplishment of CO₂R involves kinetically demanding multi-electron and multi-proton reactions (C₁, C₂₊ products) and always has competition with the thermodynamically favorable hydrogen evolution reaction (HER) (Fig. 1b) [7]. Thus, CO₂R often needs an efficient and cheap catalyst that operates at room temperature and pressure. Moreover, converting CO₂ to multi-carbon products (such as C₂H₄, C₂H₅OH, etc.) is most desirable from the industrial perspective. Still, this process always results in a mixture of products that need further downstream separation and purification costs. A perusal of CO₂R research reveals that the most crucial aspect of this endeavor is governed by catalyst development, its integration into the device prototypes, and benchmarking of the system [8–10]. Both homogeneous and heterogeneous chemistry widely pursues catalyst development for the electro or photochemically-driven CO₂ reduction [11, 12]. One of the promising classes of catalysts evolves as molecular catalysts based on earth-abundant

3d transition metal coordination complexes [13, 14]. Typically, the metal center (Fe, Co, Ni, Cu, etc.) plays the central role in binding the CO₂ molecule; the corresponding ligand framework provides a much-needed electronic environment for facile electron and proton transfer to the bound CO₂ molecule. Molecular catalysts have the most prominent advantages as they offer easy tuning of the ligand structure encompassing different electronic, steric, and electrostatic effects in the CO₂R catalyst microenvironment [8, 10, 15]. Also, they provide an adequate ligand framework to check the efficiency of different transition metals and benchmarking of catalysts for CO₂R. A handful of literature on CO₂ reduction reports various molecular catalysts based on cyclam, phosphine, carbene, bipyridine, terpyridine, quaterpyridine, porphyrin, and phthalocyanine, etc., as ligand frameworks [13]. Among them, porphyrin-based catalysts emerge as the most promising and possess the potential to address the current challenges for advancing CO₂R technology [10, 16, 17].

Porphyrins are 18π conjugated macrocyclic systems and are well-known for their rich coordination and redox chemistry [18]. Furthermore, structural modifications of typical tetrapyrrolic porphyrin give rise to versatile porphyrin analogous, which are generally termed porphyrinoids (Fig. 2a) [19].

For example, 18π electron conjugated corrole, which possesses one less *meso*-carbon than porphyrin [20], and phthalocyanine, which incorporates four conjugated isoindole units linked via four nitrogen atoms [21], etc. Metalloporphyrin-based systems are ubiquitous in biological systems, as they play crucial roles like light absorption in chlorophyll, O₂ and CO₂ transport in hemoglobin and myoglobin, electron transfer in cytochromes, etc. [18]. Inspired by the biological relevance, different metalloporphyrinoids have been explored for catalysis, light harvesting, solar cell, and photodynamic therapy applications, and CO₂ reduction studies are not excluded as well [22–24]. In the 1970s and 1980s, early electrochemical CO₂ reduction reports revealed the use of Co and Ni-phthalocyanine by Meshitsuka [25] and Fe-porphyrins by Savéant *et al.* [26]. Continuing the trend, in the last ten years, an upsurge in the reports is noted as CO₂R

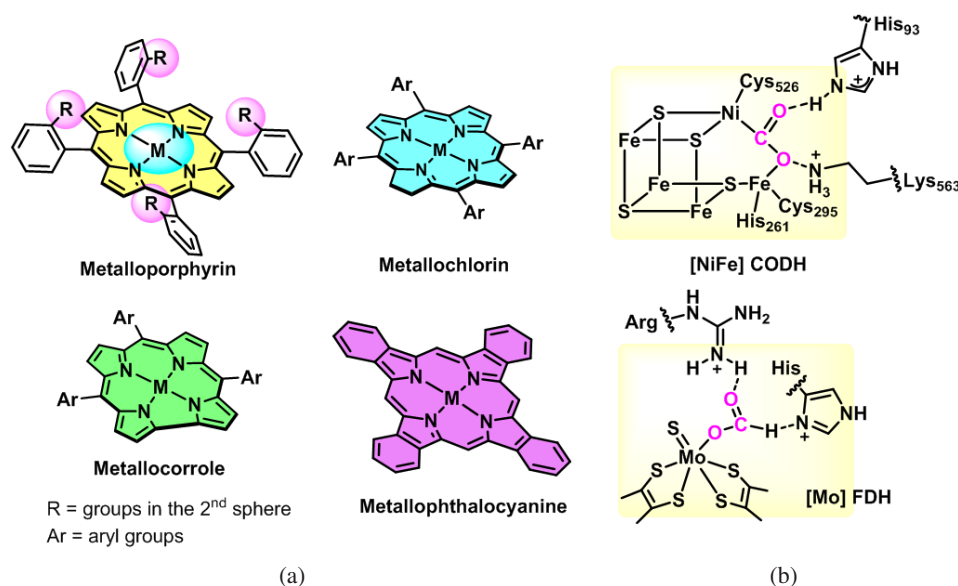


Fig. 2. (a) Molecular structure of different porphyrinoids explored for CO₂ reduction, (b) Weak interactions present in the CO₂ bound structure of active sites in [NiFe]-CODH and [Mo]-FDH enzymes, (CODH= CO dehydrogenases, FDH= Formate dehydrogenases).

becomes a countermeasure to the fossil-fuel problems. Here, an essential question arises, what makes metalloporphyrinoids so unique in the context of CO₂ reduction catalysis? Well, they possess a couple of advantages:

(1) Molecular orbital of CO₂ depicts that reduction proceeds via the electron filling in the LUMO, which is a C-O σ^* orbital. From the orbital overlap point of view, it can favorably be overlapped by the d_z^2 orbital of the metal center. Thus, the metal center (*e.g.*, Fe(0), Co(I)) with filled d_z^2 orbital in a tetragonal field (d^8 configuration) is preferred to activate CO₂. The conjugate and flexible nature of metalloporphyrinoids provides that adequate ligand field and stabilizes such metal configuration during the catalytic cycle [8]. Furthermore, metalloporphyrins are also redox-active ligands [27]; thereby, they can accumulate reducing equivalent and/or protons which are often beneficial for the reduction of CO₂. For example, redox-active porphyrinoid ligands with a redox-innocent metal center (*e.g.*, Zn²⁺) can also facilitate the CO₂ reduction [28].

(2) The bio-inspired approach for designing new CO₂R molecular catalysts is quite popular. Close inspection of the CO₂-bound adduct in the naturally available enzymes (carbon monoxide dehydrogenase (CODH) and formate dehydrogenase (FDH)) [29] reveals that the negatively charged C-O bonds are stabilized by a Lewis-acid (Fe²⁺) center and/or by H-bonding through nearby amino acid residues (Fig. 2b) in an aqueous environment with relatively low reorganizational energy. A similar microenvironment is rationally mimicked around the central metal atom of porphyrinoids by employing suitable substituents in the *meso*-aryl groups bearing electronegative, H-bonding, and columbic effects, etc. For example, A₄, A₂B₂, or A₃B types of porphyrinoids are synthesized *via* well-established synthetic protocols with the desired tuning of electronic and spatial properties for the CO₂R study.

(3) Metalloporphyrinoids are air-stable and cheap compared to the Nobel metal CO₂R catalysts (*e.g.*, Au, Ag, etc.). It is susceptible to mild to relatively harsh reaction conditions such as metalation, hydrothermal, formylation, click reaction, Schiff base condensation, etc. [30]. Therefore, it expands the scope of testing porphyrinoids in combination with other solid materials ranging from meso to nanoscale dimensions.

(4) Owing to the distinctly visible region absorptions, metalloporphyrins show signature absorption, fluorescence, excited state dynamics, and vibrational characteristics in different metal oxidation states [31]. Such intriguing spectroscopic properties of metalloporphyrinoids put them at the forefront while probing CO₂ reduction mechanisms by various sophisticated in-situ and ex-situ spectroscopy analyses. Mechanistic understanding is key to evaluating catalyst degradation and structure-activity relationship, and it can be deciphered using the lucrative properties of the porphyrinoids.

In the past two years, several well-documented reviews appeared covering broader scopes and different aspects of metalloporphyrins for CO₂ reduction [8, 10, 16, 17, 32]. However, they mainly covered metal complexes of porphyrins (Por) and phthalocyanines (Pc), although, in recent times, distinct progress has been achieved with metal complexes of corrole (Cor) and chlorin (Chl) type porphyrinoids. CO₂ reduction is also a hot topic; new efficient systems are rapidly evolving in short periods. Against this backdrop, the current review focuses on the recent advances in homogenous and heterogenous CO₂ reduction with different metalloporphyrinoid catalysts. Herein, we have summarized the emerging CO₂ reduction catalysts based on their rational design perspective. We have also highlighted the key parameters such as electrode materials, overpotential (η), Faradaic efficiency

Homogeneous medium	Heterogeneous medium	Testing in electro/photoreactors
Mode of CO₂ reduction study Electrochemical (EC) Photochemical(PC) Mainly single compartment cell	Electrochemical (EC) Photochemical(PC) Photoelectrochemical (PEC) Single and H type cell	Flow cell electrolyze (FCE) Flow cell photoreactor (FCP) Photovoltaics-electrochemical (PV-EC)
Catalyst design 2 nd sphere modification: (H-bonding, columbic interaction, proton source) Extension of conjugation Partially hydrogenated porphyrins Bimetallic structure	Composite material Covalent anchoring COF, MOF or coordination polymer Electrodeposition Mixing with nanomaterials (E.g., Cu NPs)	Coating on Gas Diffusion electrode (GDE) Composite materials Anchoring strategy Bottom Up catalyst integration
Key points Gain in overpotential value In-situ and ex-situ operando spectroscopy Mechanistic study	Gain in overpotential, specially in PEC Enhanced route to C ₂ and C ₃ product Study in bicarbonate solution and stability	High current density ($\geq 100 \text{ mA cm}^{-2}$) Industrial feasibility (Long term stability) TRL and CAPEX analysis

Fig. 3. State-of-the-art approaches for the designing of the metalloporphyrinoids-based CO₂ reduction catalysts.

(FE), product selectivity, turnover numbers (TON), and turnover frequencies (TOF) as reported.

State-of-the-art approaches

Rational designing of the catalyst ranges from molecular structure level to supported composite material or polymeric materials. The decorum is set to achieve the best key CO₂R metrics and obtain the ‘Holy Grail’ of CO₂ reduction catalysis. Accordingly, we have divided the catalyst developments into three broad categories (Fig. 3).

Homogeneous medium: Both electro and photochemical CO₂ reduction is performed in the homogeneous condition. It is mainly driven by bio-inspired approaches relying on installing weak interactions in the catalytic micro-environment [10]. Here, structural modification of the porphyrinoids is carried out to alter the electronic and steric properties of the molecule. The *meso*-aryl substituents and the axial position of the central metal atoms are judiciously utilized to incorporate specific functional group(s) or atoms. They have often been tested in organic medium and are suitable for details mechanistic analysis [33]. Nevertheless, with this approach, producing CO₂ reduction beyond 2 electron products (formic acid or CO) is rare.

Heterogeneous medium: Metalloporphyrinoids exhibited excellence in organic medium, but it remains poorly soluble in water medium. For large-scale CO₂R applications, the aqueous medium is the only way to go, thereby putting metalloporphyrinoids on solid support before studying catalysis becomes necessary. The molecular complexes adsorbed onto or covalently bound to the electrode surface as a film and explored for electro, photo, and photoelectrochemical CO₂ reduction. As a conductive support material, metal nanoparticles (Cu, Ag, etc.), carbon materials (Vulcan carbon, multiwalled carbon nanotubes (MCNT), and the graphene), or solid photosensitizers (e.g., TiO₂, GaP, ZnO, semiconductors) are used to augment the performance of the molecular catalyst. The incorporation of metalloporphyrinoid units in reticular materials (MOFs, COFs) or coordination polymers is also carried out and tested for the CO₂ reduction

[34, 35]. During the material fabrication, acquired knowledge of homogeneous catalysis is also utilized, and a similar boosting effect is anticipated during the catalysis. For example, post-synthetic modifications on the MOFs, and COFs are carried out by adding favorable functional groups in the vicinity of the metalloporphyrinoid unit [36]. A tandem approach is also evolved in which the first CO₂ is reduced to a C₁ product (e.g., CO) and then further *in-situ* reduction of the C₁ product into C₂ or C₃ (liquid/gaseous) products [37, 38]. Here, combining two CO₂R materials to have such transformation becoming popular and heterogeneous catalysts based on metalloporphyrinoid are also explored. However, the reactivity and CO₂ product selectivity are controlled by the morphology, structure, pore size, and band structure of the overall materials. As a result, in this case, low partial current densities and selectivity for the C₂₊ products have often been observed.

In Electro and Photoreactor: Escalating advancement in the homogenous and heterogeneous CO₂ reduction opened up possibilities to build an actual device for CO₂ electro/photolysis in the market, and that’s the supreme goal of current CO₂R research. Interestingly, CO₂ to CO electrolyzers with metalloporphyrinoids have already been tested from lab-scale to pilot-scale setups [39]. The electrolyzers are constructed using flow-cell assemblies where CO₂R occurs in a gas-liquid-solid interface [5]. In this case, the metalloporphyrinoids are deposited on a gas diffusion electrode (GDE) made of carbon materials. Catalysts integration in CO₂ reactors currently follows similar approaches to heterogeneous ones, but it has plenty of scope to improve.

In the following three sections, we will briefly discuss the different strategies for metalloporphyrinoid catalyst fabrication according to these three categories and their CO₂ reduction performances.

Metalloporphyrinoids in homogeneous CO₂ reduction

Significant advances in electro and photochemical CO₂ reduction in the homogeneous medium are depicted with metalloporphyrins, and a few sporadic reports on

metallocorroles and metallochlorins have also emerged. However, solubility issues associated with metallophthalocyanines have forced them to be mainly explored in the heterogenous medium, which is described later.

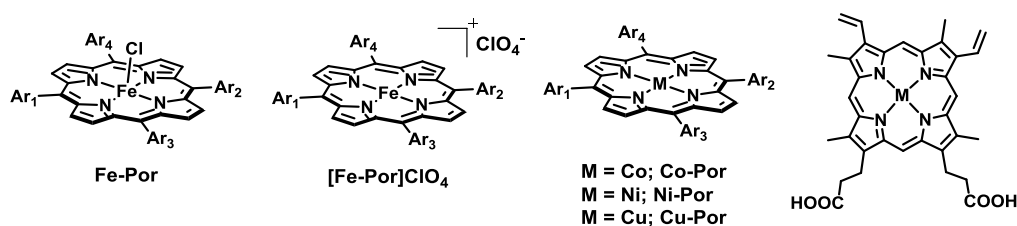
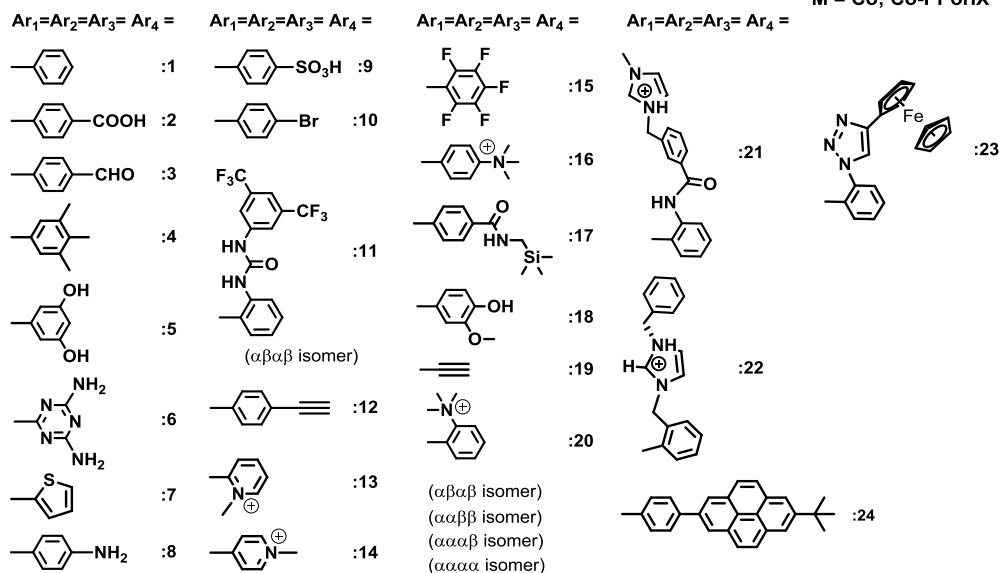
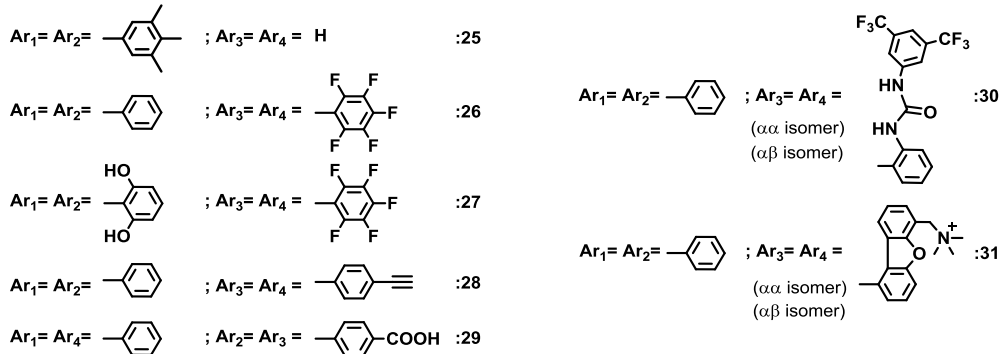
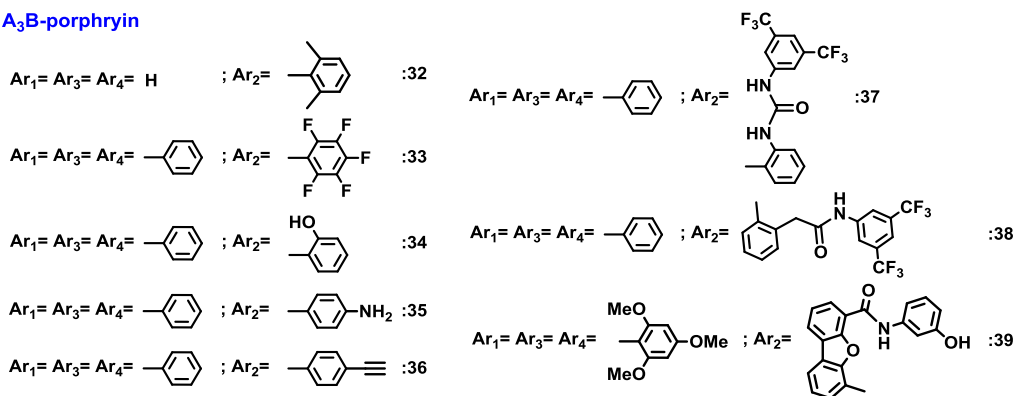
Metalloporphyrins

Iron porphyrin is the most studied congener out of all the metalloporphyrinoids [40]. In general, the cyclic voltammogram of iron porphyrin showed three reversible metal-centered reduction peaks ($\text{Fe}^{\text{III/II}}$, $\text{Fe}^{\text{II/I}}$, $\text{Fe}^{\text{I/0}}$) in non-aqueous solvents. The CO₂ activation by Fe porphyrin reckons on a *push-pull* type mechanism. The formal Fe⁰-porphyrin *pushes* the electrons to the CO₂ and forms a negatively charged adduct ($[\text{Fe}^{\text{I}}(\text{CO}_2)]^{2-} \rightleftharpoons [\text{Fe}^{\text{II}}(\text{CO}_2)]^{2-}$) [41]. The adduct further undergoes C-O bond cleavage by protonation and intramolecular electron transfer to produce Fe^{II}-CO intermediate. Later, CO is released by reductive decarbonylation (by another Fe⁰ species) and regenerates the catalyst. Brønsted or Lewis acid species realize the *pull effect*, which accelerates the C-O bond-breaking step and product formation [42, 43]. Savéant and co-workers proposed this mechanism based on the simulations of electrochemical parameters and cyclic voltammetry responses [44, 45]. However, no such intermediate was isolated or spectroscopically characterized until Dey and co-workers successfully characterized two intermediates ($[\text{Fe}^{\text{II}}(\text{CO}_2)]^{2-}$, Fe^{II}-COOH) involved in the proposed cycle by resonance Raman and DFT calculations [46]. However, their study was specific to the use of particular *meso* substituents of the Fe-porphyrin, solvent, proton source, and temperature. Currently, plenty of scopes are available in the elucidation of porphyrin-assisted CO₂ reduction mechanisms supported by various spectroscopic techniques. Some important recent reports on this note will be discussed later, along with the specific metalloporphyrinoid examples. Nevertheless, for our current discussion, we will concentrate on the rational designing principle of the Fe porphyrins, which is mainly governed by using the functional groups in the vicinity of the Fe center. The desired functional groups help in stabilizing the initial negatively charged adduct ($[\text{Fe}^{\text{II}}(\text{CO}_2)]^{2-}$) and further facilitate the proton and electron transfer process for easy C-O bond cleavage. In the last ten years, investigations on the suitable substituents effects on iron porphyrin have been guided by two routes: through-structure (electronic, inductive) effects and through-space (H-bonding, columbic interaction) effects. Through-space effects are proven much more effective for kinetic factor gain (high TON, TOF) during CO₂ reduction, while through-structure tuning helps to attain thermodynamic gain (overpotential). Herein, the designing of porphyrins is envisaged via *Second sphere tuning by changing the meso-aryl substituents* so that both the thermodynamic and kinetic factors are optimally available. Savéant and co-workers reported the systematic effect of the electron-withdrawing (pentafluorobenzene),

proton donor (phenolic -OH), or positively charged (trimethyl ammonium group at *ortho/para* position of the *meso*-phenyl ring) substituents with **Fe-Por-5**, **Fe-Por-2**, **Fe-Por-15**, **Fe-Por-9**, **Fe-Por-26**, **Fe-Por-27**, **Fe-Por-16**, **Fe-Por-20**, etc. by comparing iron-tetraphenylporphyrin **Fe-Por-1** as standard [47–50]. Now, standing on this sound background, we will focus on the recent results obtained with various iron porphyrins for CO₂ reduction.

H-bonding, proton source, and amine precursors:

Chang and co-workers reported a functional mimic of the [NiFe]-CODH with several Fe porphyrins, bearing hydrogen-bond and proton relay groups for the electrochemical CO₂R. They have synthesized a series of A₃B type *ortho*- or *para*- substituted amide pendants at the periphery of the iron porphyrin [51]. The *ortho*- amide isomer **Fe-Por-38** exhibited the best performances for CO formation ($\text{FE}_{\text{CO}} = 92\%$ and $\text{TOF} = 5.5 \times 10^6 \text{ s}^{-1}$) in DMF containing 100 mM phenol as it was engaged in favorable through-space interaction during the catalytic cycle. Later, they prepared another series of iron porphyrins decorated with imidazolium pendants as multifunctional charge and hydrogen-bond donor groups [52]. **Fe-Por-22** turned out as the best candidate for the CO₂ to CO conversion with a 25-fold increase in the CO₂ binding affinity and 2000-fold CO₂R activity compared to the benchmark **Fe-Por-1** catalyst. In this case, the synergistic through-space electrostatic and H-bonding interactions are responsible for the improved performance. Most recently, using **Fe-Por-37** in DMF containing tetraethyl ammonium bicarbonate, they have adopted a bicarbonate templating strategy in the vicinity of the Fe-center using the urea pendant present in the second coordination sphere [53]. The bicarbonate template acts as a viable proton donor in DMF electrolyte ($\text{pK}_a = 20.8$) and resulted in a 1500-fold increase in the CO₂R efficiency ($\text{FE}_{\text{CO}} = 98\%$) compared to the unmodified **Fe-Por-37**. Aukauloo and co-workers also reported several *meso* tetra-aryl substituted iron porphyrins relying on the ionic liquid environment or multipoint hydrogen bonding groups in the second coordination sphere. First, they reported a water-soluble iron porphyrin **Fe-Por-21** containing methylimidazolium groups as ionic liquid units [54]. **Fe-Por-21** produced CO ($\text{FE}_{\text{CO}} = 91\%$) in DMF/water (9:1) electrolyte at -0.948 V vs. NHE with a TOF value of $2.44 \times 10^5 \text{ s}^{-1}$. Later, they reported another iron porphyrin (**Fe-Por-11**) containing four urea and amide functions in $\alpha\beta\alpha\beta$ atropisomeric form [55]. **Fe-Por-11** exhibited better CO₂R reactivity with CO formation at $\sim 300 \text{ mV}$ more positive potential in DMF electrolyte when compared to analogues **Fe-Por-1**. Furthermore, water entrapped in the porphyrin cleft was found a sufficient and a better proton source for the catalytic cycle in comparison to more acidic and commonly used trifluoroethanol and phenol. In another report, they modified the same iron porphyrin structure and synthesized an A₂B₂ type iron porphyrin (**Fe-Por-30**) containing only two urea functions [56]. Out of the both possible $\alpha\alpha$ and $\alpha\beta$ atropoisomers of **Fe-Por-30**, the $\alpha\alpha$ isomer

**A₄-porphyrin****A₂B₂-porphyrin****A₃B-porphyrin****Scheme 1.** Metalloporphyrins used for CO₂ reduction catalysis.

showed stronger binding affinity towards the incoming CO₂ molecule by the existing a tight water hydrogen-bonding network whereas the $\alpha\beta$ isomer exhibited better electrocatalytic activity for CO₂ to CO formation in DMF containing water as a proton source. The higher catalytic activity for $\alpha\beta$ isomer ($K_{\text{cat}} = 5.14 \times 10^4 \text{ s}^{-1}$) over $\alpha\alpha$ isomer ($K_{\text{cat}} = 1.21 \times 10^4 \text{ s}^{-1}$) was attributed to free access of the protons to the CO₂ bound Fe-center on the opposite side of the Urea unit upon addition of stronger acid. This demonstrates that the different hydrogen bonding strength in two atropisomers of an iron porphyrin imparts an effect on the CO₂ binding and catalytic reduction rates. Nocera and co-workers reported a series of A₃B type iron hangman porphyrins with phenol, guanidinium, and sulphonic acid proton donor groups suitably placed above the Fe-center [57]. As one of the examples, in **Fe-Por-39**, the proton relay effect of the phenolic -OH group accelerated the CO₂ binding within the hangman cleft and showed excellent CO₂ to CO conversion ($\text{FE}_{\text{CO}} > 93\%$). Welch and co-workers reported *meso*-tetrathienyl iron porphyrin **Fe-Por-7** which showed comparable FE_{CO} and TON_{CO} relative to the benchmark **Fe-Por-1** catalysts [58]. However, **Fe-Por-7** exhibited a 150 mV lowering in the overpotential as thiophene rings imparted effective extension of π -conjugation of the porphyrin, leading to stabilization of the LUMO energy level.

Additionally, electrochemical CO₂ to CO formation was directed to formate (selectivity up to 70%) by using the same **Fe-Por-1** upon the addition of a tertiary amine (e.g., quinuclidine) in presence of a weak proton donor [59]. The basic amine axially coordinated to CO₂ bound Fe-center and it pushed more electron (*trans* effect) density to the bound 'C' atom. It further stabilizes the intermediate (Fe(II)-COOH) and facilitates formate formation over CO.

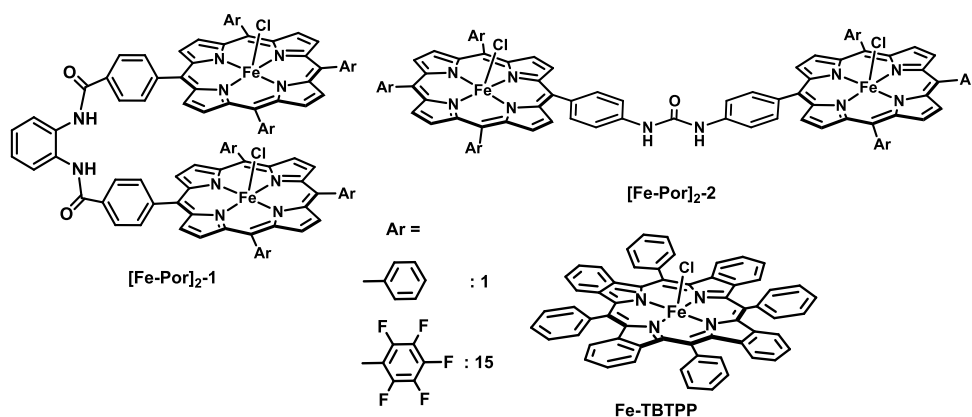
Electrostatic interactions: Savéant and co-workers reported **Fe-Por-20** bearing four positively charged *ortho*-trimethyl ammonium groups, which is one of the best iron porphyrins for electrochemical CO₂ to CO conversion in DMF solvent [47]. Authors reported the CO₂R performance of **Fe-Por-20** with its $\alpha\beta\alpha\beta$ atropoisomeric form, although typical porphyrin condensation resulted in the formation of other three atropisomers ($\alpha\alpha\alpha\alpha$, $\alpha\alpha\beta\beta$, $\alpha\alpha\alpha\beta$). Moreover, all these atropisomers are easily interconvertible in solution at room temperature and a mixture of them is anticipated to contribute differently toward the CO₂R performances. Mayer and co-workers recently investigated this aspect and showed the effect of the orientation of the positively charged *ortho*-trimethyl ammonium groups on the CO₂R performances with four such iron porphyrin **Fe-Por-20** atropisomers individually [60]. Based on Cyclic voltammetry experiments in DMF and phenol as a proton source, the best performance was obtained with the $\alpha\alpha\alpha\alpha$ isomer ($\text{TOF} = 2 \times 10^5 \text{ s}^{-1}$ at $\eta = 0.48 \text{ V}$), although other isomers showed comparable results. This study confers that catalytic benefit in **Fe-Por-20** family arises due to the presence of the net overall positive charge in the vicinity of the metal center during

the CO₂R, not from the orientation of the individual positive charges. Cao and co-workers recently reported *trans*-A₂B₂ type Fe porphyrin **Fe-Por-31** with two cationic trimethylbenzylamine groups [61]. The $\alpha\alpha$ isomer of **Fe-Por-31** showed larger activity than the $\alpha\beta$ isomer with a $\text{TOF} = 4.4 \times 10^5 \text{ s}^{-1}$ in acetonitrile and phenol as a proton source. Theoretical studies further unveil that cationic units not only help in the stabilization of the negatively charged Fe-CO₂ adduct but also promotes C-O bond cleavage through the role-specialized cooperation effect.

Recently, A. Khadraoui *et al.* also studied the through-space electrostatic effect using a series of tetra-, di-, and mono-substituted iron porphyrins bearing cationic imidazolium groups [62]. It was reported that iron porphyrins with fewer imidazolium units showed increasing overpotential values ranging from 230 to 430 to 620 mV, compared to standard iron-tetraphenylporphyrin **Fe-Por-1**. The mono-substituted iron porphyrin exhibited the best turnover frequencies with a $\text{TOF} = 2.05 \times 10^8 \text{ s}^{-1}$ and $\text{FECO} = 89\%$.

O₂-tolerant CO₂ reduction: In the flue gases, the presence of O₂ is a considerable amount along with CO₂, hence, successful CO₂R technologies need O₂-tolerant catalysts. For such O₂-tolerant CO₂R, Dey and co-workers reported a ferrocene appended Fe porphyrin **Fe-Por-23** type bi-functional catalyst [63]. **Fe-Por-23** facilitates selective electrochemical CO₂ to CO reduction *via* the central Fe(0) center whereas four-electron reduction of the O₂ to H₂O proceeds through the three of the four ferrocenes are oxidized along with the Fe(II) center of the porphyrin.

Solvent and counter anion effect with Fe-porphyrin: Apart from extensive efforts in the structural tuning of the Fe-porphyrin, of late studies have been carried out on the solvent effects for the CO₂ reduction activity by Fe-porphyrins. In general, DMF is considered a suitable solvent for studying the catalytic activity of metalloporphyrins mainly due to its easy solubility factor. However, the solubility of CO₂ gas is slightly greater in acetonitrile (0.28 M) than in DMF (0.20 M) [64]. Bin *et al.* systematically examined the CO₂ reduction activity of porphyrin **Fe-Por-15** in acetonitrile, propylene carbonate, benzonitrile, and THF [65]. Although, a previous study on **Fe-Por-15** in DMF showed poor activity for the CO₂ to CO conversion [48]. Porphyrin **Fe-Por-15** displayed the highest activity in acetonitrile ($\text{TOF} = 4.2 \times 10^4 \text{ s}^{-1}$), and had modest activity in propylene carbonate. Factors like coordinating ability, polarity, and dielectric properties of the respective solvents cumulatively contributed towards the observed activity difference of **Fe-Por-15** in the different solvents. Warren and Sinha reported a monosubstituted unsymmetrical (A₃B) Fe porphyrin **Fe-Por-34** that also displayed higher CO₂ to CO reduction activity (100-fold enhancement in TOF) in acetonitrile than in DMF [66]. The activity difference originates from the different H-bonding ability of acetonitrile and DMF molecules and the subsequent effect of their H-bonding interaction with the *meso*-2-hydroxyphenyl group which alters the key proton relay process during the CO₂ reduction. In



Scheme 2. Bimetallic Fe-porphyrins and Fe-tetrabenzo-tetraphenylporphyrin were reported for CO₂ reduction catalysis.

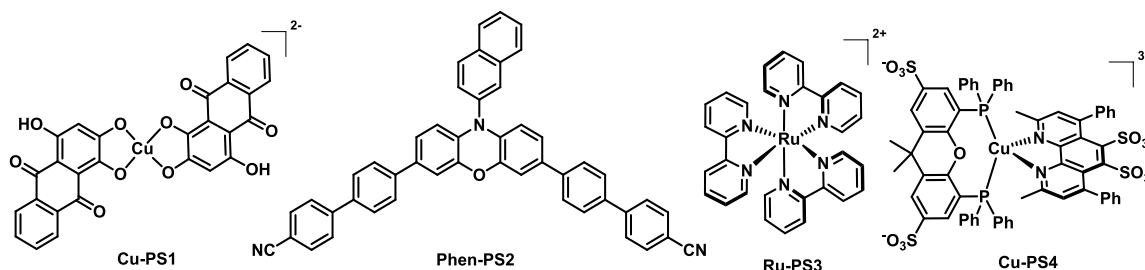
2021, Kondo, Masaoka group and co-workers reported an easy tuning of the Fe porphyrin structure which dramatically increase its electrochemical CO₂ to CO reduction activity in acetonitrile solvent (TOF = 7.3×10^6 s⁻¹) [67]. The typical iron(III)tetraphenylporphyrin chloride **Fe-Por-1** (TOF = 1.1×10^2 s⁻¹ in DMF) is subjected to an ion exchange process and converted into a cationic complex **[Fe-Por]ClO₄-1** with ClO₄⁻ as counter anion before studying in acetonitrile (Scheme 1). CV and UV-visible absorption SEC measurements suggested that in acetonitrile, CO₂ reduction catalysis with **[Fe-Por]ClO₄-1** followed at the Fe^I/Fe^{II} redox couple (thereby lower overpotential value) in contrast to typical Fe⁰/Fe^I based CO₂ activation in DMF. Thus, these recent studies highlight the inevitable role of the solvents and invoke consideration of the solvent role while comparing the catalytic efficiencies of Fe porphyrins.

Bimetallic Fe-porphyrin: Naruta and co-workers explored a series of co-facial Fe porphyrin dimers for CO₂ reduction with the aim to have metal atoms co-operativity effect during the CO₂ binding and C-O bond cleavage process [68, 69]. Their recent report revealed the use of co-facial Fe porphyrin dimer with a flexible o-phenylene amide bridge **[Fe-Por]₂-1-15** for CO₂ to CO conversion both in non-aqueous and aqueous media (Scheme 2) [70]. In DMF in the presence of 10% water/1 M Phenol as proton donor **[Fe-Por]₂-1-15** produced CO at η value of 0.15 V with 94% FE_{CO}. For the aqueous medium, **[Fe-Por]₂-1-15** was immobilized to ketjan carbon black and grafted on a glassy carbon electrode, and in this case,

production of CO with 12.8 mA.cm⁻² current densities at η value of 0.46 V was reported.

Photocatalytic CO₂ reduction in a homogeneous medium essentially comprises three components, *i.e.*, a molecular catalyst, a redox photosensitizer (PS), and a sacrificial electron donor (*e.g.*, triethaloamine). Light-driven electron transfer occurs from the photosensitizer to the metal complex which in turn accumulates multiple electrons needed to pursue CO₂ reduction. In addition, photocatalytic CO₂ reduction performance is often enhanced by using a proton donor (weak acids, *e.g.*, tri-fluoroethanol) as an extra component. Classically, Ru or Ir bipyridines (*e.g.*, **Ru-PS3** in Scheme 3) with lower reduction potential are used as photosensitizers for CO₂ reduction using metalloporphyrins as catalysts.

However, currently, major research is going on the development of low-cost photosensitizers (with 3d transition metal) or organic photosensitizers. With recent advances, few reports are evolved describing a range of organic dye or inorganic metal-based photosensitizers (Scheme 1) with Fe or Co porphyrins as catalysts. Han's group showed dramatic improvement in photocatalytic CO production (95% selectivity over H₂, TON = 16100, TOF = 7650 h⁻¹) using **Fe-Por-5** as a catalyst, copper purpurin **Cu-PS1** as photosensitizer (Scheme 3) and 1,3-dimethyl-2-phenyl-2,3-dihydro-1H-benzo[d]imidazole (BIH) as the sacrificial reductant in DMF [71]. The same **Fe-Por-5** with Ir(ppy)₃ showed very weak performance for the CO production (TON = 140 in 55 h) as previously studied [72]. Mechanistic analysis showed



Scheme 3. Molecular structure of photosensitizers used for photochemical CO₂ reduction.

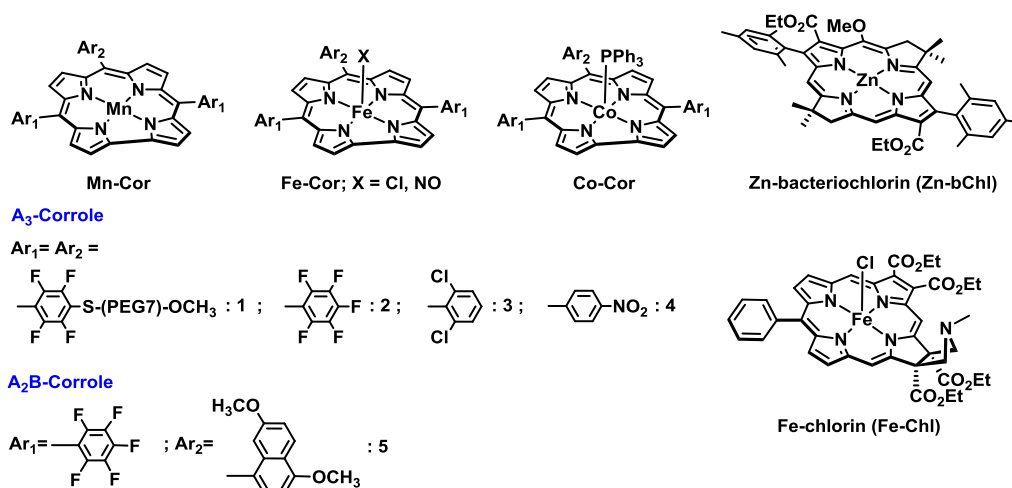
that Cu complexation to the purpurin ligand (**Cu-PS1**: -1.75 V vs. SCE) resulted in a shift of 540 mV with more negative reduction potential than the organic dye purpurin (-1.21 V vs. SCE). The drastic negative reduction potential drives the electron transfer to the active Fe(0) form of **Fe-Por-5** (Fe^{II}: -1.55 V vs. SCE) catalyst at a faster rate, thereby enhanced CO₂ reduction is observed. Martin *et al.* evaluated several phenoxazine derivatives as visible light photosensitizers with **Fe-Por-16** as catalysts and BIH as a sacrificial electron donor [73]. The best performance was obtained with **Phen-PS2** (Scheme 3) in acetonitrile with 100% CO selectivity although the turnovers (TON = 115) remained significantly low. Production of CO was proved to depend on the oxidation potential of phenoxazines and their regeneration through the electron transfer from the BIH during the catalytic cycle. Aukaaloo *et al.* also explored the urea-functionalized porphyrin **Fe-Por-11** for the photochemical CO₂ reduction in DMF medium [74]. Under blue LED irradiation ($\lambda > 460$ nm), **Fe-Por-11** in DMF/water (9:1) solution in presence of **Ru-PS3** as photosensitizer and BIH as sacrificial electron donor produced CO in with 99% selectivity with a TON of 7271 and TOF of 3720 h⁻¹. Most interestingly, they characterized the CO₂ activation by the Fe(I) state of **Fe-Por-11** during the photocatalytic process, assisted by the urea functional groups. This is one of the first reports which invokes the existence of an alternative mechanism of CO₂ activation by the Fe porphyrin without involving the classical Fe(0) state.

Sakai and co-workers investigated photochemical CO₂ reduction in water medium extensively using different water-soluble cobalt porphyrins [75–78]. CO₂ to CO conversion in fully aqueous media (pH 6.7) was achieved with **Co-Por-9** (82% selectivity, TON = 926) and **Co-Por-2** (74% selectivity, TON = 2500) porphyrin in the presence of [Ru(bpy)₃]²⁺ **Ru-PS3** as photosensitizer and ascorbate as sacrificial electron donor (Scheme 1). In both cases, Co(I) is the active catalyst form to bind CO₂ and initiate the catalytic cycle. Progressive degradation of [Ru(bpy)₃]²⁺

was identified as a limiting factor for the lowering of the overall photocatalysis rate. Later, they developed a highly luminescent water-soluble heteroleptic copper (I) diphosphine and diimine-based complex **Cu-PS4** as photosensitizer (Scheme 3). With **Cu-PS4** as a photosensitizer, three cobalt porphyrins **Co-Por-2** (90% selectivity, TON = 1085), **Co-Por-13** (90% selectivity, TON = 4000), and **Co-Por-14** (77% selectivity, TON = 2680) fetched CO as the main product (Scheme 1). The enhanced photocatalysis in the case of N-methyl pyridinium cobalt porphyrins (**Co-Por-13** and **Co-Por-14**) was attributed to the reduction at the four pyridinium moieties along with the Co center and the overall 4 or 5 electrons reduced electron-rich porphyrin, which triggers efficient CO₂ binding.

Metalloporroles

Gross and Fujita's group first reported the electro and photochemical CO₂ reduction activity of Fe and Co corroles [79]. Electrochemical CO₂ reduction to CO in acetonitrile solution was reported with 5,10,15-tris-pentafluorophenyl cobalt (III) **Co-Cor-2** and iron (IV) **Fe-Cor-2** corroles (Scheme 4). Another iron corrole **Fe-Cor-3** bearing three 2,6-dichlorophenyl groups was also subjected to the same CO₂ reduction activity (Scheme 4). Cyclic voltammetry analysis revealed that CO₂ binds with the Co(I) and Fe(I) corroles, unlike the case of the respective porphyrins and phthalocyanines, where Co(0) or Fe(0) states are active for the CO₂ reduction. Moreover, Fe corrole **Fe-Cor-3** was proven as the most efficient electrocatalyst. Later, Sinha *et al.* reported electrochemical CO₂ reduction to CO by iron **Fe-Cor-5** and cobalt **Co-Cor-5** unsymmetrical A₂B corroles bearing 4,7-dimethoxy naphthaldehyde and pentafluorophenyl groups (Scheme 4) [80]. The study revealed that the (nitrosyl)iron complex (**Fe-Cor-5**) performs much better catalysis than the cobalt complexes **Co-Cor-5** and the (chloro)iron complex **Fe-Cor-3**. Photochemical CO₂ reduction was also reported with iron corrole **Fe-Cor-3**



Scheme 4. Metalloporroles and Metallochlorins are used for CO₂ reduction catalysis.

in acetonitrile solution containing *p*-terphenyl (TP) as a sensitizer and triethylamine (TEA) as a sacrificial electron donor (Scheme 3). Under the photochemical condition, the production of CO and H₂ proceeded via the Fe(I) oxidation state [79].

Metallochlorins

Under electro and photochemical CO₂R condition, irrespective of the metal centers, porphyrin ligands undergo degradation due to irreversible hydrogenation of the double bonds in pyrrole rings, forming phlorin or chlorin-type structures [81]. Thus, using chlorin metal complexes from the start is expected to show better CO₂ reduction efficiency as the chlorins are stable towards unwanted dehydrogenation pathways. In this regard, Jiang *et al.* reported a Zn(II)bacteriochlorin (**Zn-bChl**) which showed 92% CO₂ to CO Faradic efficiency under electrochemical CO₂ reduction at -1.9 V vs. Ag/AgCl (Scheme 4) [28]. The **Zn-bChl** was found stable under the reductive condition compared to its porphyrin analogue and exhibited a j_{CO} of 2.3 mA.cm⁻². DFT and CV studies suggested that CO₂ reduction proceeds via double reduction of **Zn-bChl** and CO₂ bind to the redox-innocent Zn⁺² metal center while the reducing electrons are delocalized on the redox-active bacteriochlorin ring. This is a unique example of redox noninnocence of porphyrinoid ligand dictating the electrochemical CO₂ reduction. Recently, the Dey group reported another iron chlorin (**Fe-Chl**) with a pendant amine second sphere residue which reduces CO₂ to HCOOH under the electrochemical condition in DMF containing 1M H₂O as a proton source (Scheme 4) [82]. In the case of **Fe-Chl**, the product selectivity towards HCOOH (97% selectivity) and overpotential value (50 mV) achieved is quite remarkable when compared to common Fe porphyrins, for which CO is detected as the only product and the best catalysts, **Fe-Por-22**, is active at 220 mV overpotential [83]. FTIR, resonance Raman and Mossbauer spectroscopic measurements suggested binding of CO₂ to the Fe(I) state of **Fe-Chl**. Moreover, closely placed proton transfer residue (protonated pendant amine unit) was proven to help in stabilizing the intermediate (Fe(II)-COOH) which further hydrolyses to yield HCOOH.

Metalloporphyrinoids in heterogeneous CO₂ reduction

Although homogeneous catalysis offers good control over the design of metalloporphyrinoids by structural tuning, their use in solution is associated with several drawbacks, such as limited solubility in aqueous media only a few active catalyst molecules within the diffusion layer, etc. Moreover, freely diffusing catalyst crossovers between the cathode and anode prevent effective separation of the two half-reactions. In this regard, the immobilization of molecular catalysts on the electrode surface offers the possibility of overcoming these

drawbacks. Immobilization of metalloporphyrinoids involves numerous strategies and recent advances can be classified into the following approaches: (1) covalent or non-covalent immobilization on the carbon electrode, (2) immobilization on the high surface areas of carbon support materials such as graphene or CNTs, (3) encapsulation into polymers, (4) encapsulation into reticular 3D materials which include metal-organic frameworks (MOFs), covalent organic frameworks (COFs) or covalent organic polymers (COPs) and (7) functionalization on semiconductor systems such as metal nanoparticles (NP) or metal oxide (*e.g.*, TiO₂). Heterogenized metalloporphyrinoids are explored for electro and photoelectrochemical CO₂R in CO₂-saturated NaHCO₃ medium and are often tested in two-compartment H-type cells. In this section, we will discuss the recent advances in the heterogenization of metalloporphyrins and metallophthalocyanines for CO₂R by considering each type of immobilization strategy.

Covalent or non-covalent immobilization on the carbon electrode

Covalent and non-covalent immobilization of the molecular catalysts on the carbon electrode is popular, although direct immobilization without any support material does not provide high catalytic current density. Hence, these strategies are not preferred for practical device applications and are limited to basic mechanistic studies. For covalent anchoring of molecular catalysts on a carbon electrode, copper(I)-catalyzed azide-alkyne cycloaddition or “click chemistry” has been employed successfully in recent times. Yao *et al.* reported attachment of a **Co-Por-12** on the boron-doped, p-type conductive diamond, where the peripheral acetylene group of **Co-Por-12** was covalently linked on diamond surface decorated with alkyl azides (Scheme 1) [84]. Further, **Co-Por-12** modified diamond surface was used as a ‘smart’ electrode for electrochemical CO₂ to CO conversion in acetonitrile solution. On the other hand, in the non-covalent category, Rivera-Cruz *et al.* deposited **Co-Pc-1** complexes on edge plane graphite electrodes with various pyridyl moieties which were axially coordinated to the Co center. It was found that increasing the σ -donor strength of the pyridyl ligands resulted in the enhancement of the catalytic activity of the resulting Co-Pc-(pyridyl) catalysts (Scheme 5) [85]. It has been hypothesized that axial coordination leads to an increase of the Co dz² orbital energy, which further increases the complex nucleophilicity and facilitates CO₂ coordination compared to the parent **Co-Pc-1** complex. Sinha *et al.* chose basal plane graphite (BPG) electrodes to immobilize **Co-Por-1** and **Co-Por-34** by simply drop casting method and constructed **Co-Por-1/BPG** and **Co-Por-34/BPG** films, respectively (Scheme 1). Electrochemical CO₂R study using **Co-Por-34/BPG** produced mainly CO with FE_{CO} = of 92% at -0.85 V vs. NHE, which has

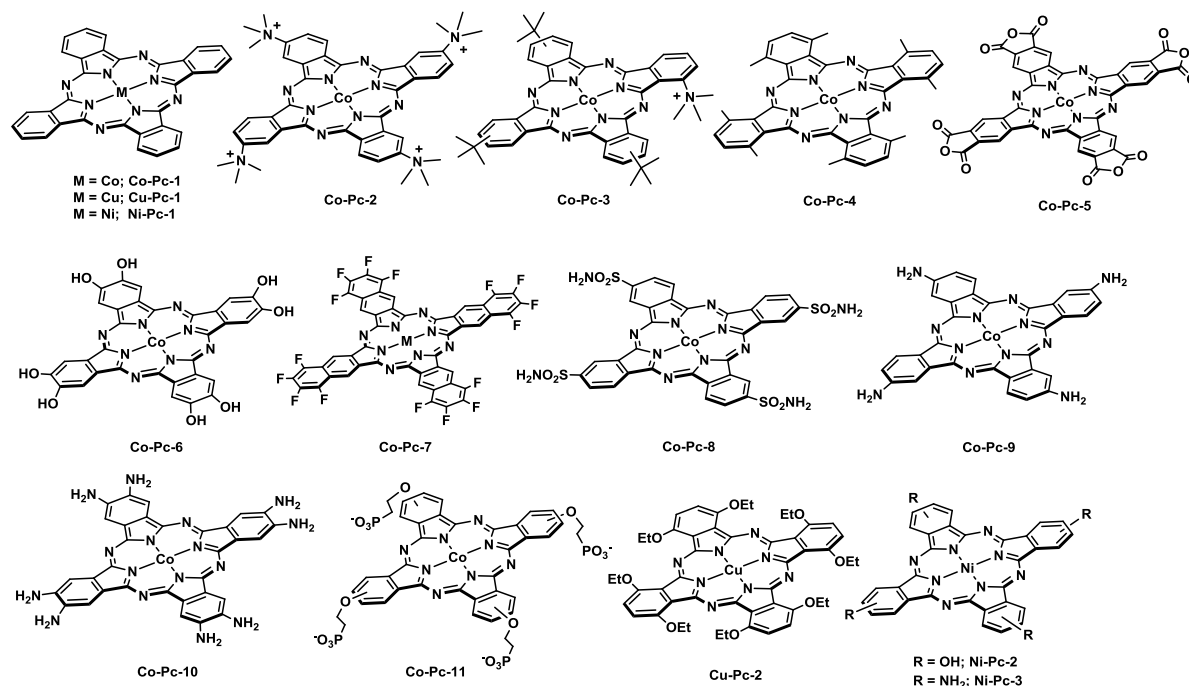
a gain of 160 mV in overpotential when compared to **Co-Por-1**/BPG. It was evinced that the 2-hydroxyphenyl group on the porphyrin **Co-Por-34** significantly enhances CO₂ reduction also in the heterogeneous phase due to a single proton relay from the ligand in the aqueous electrolyte [86].

Immobilization with carbon support materials

High surface area carbon materials, such as graphene [87] and carbon nanotubes [88] are regularly used as molecular catalyst supports because of their good conductivity, durability, and aromatic nature with delocalized π electrons that facilitate rapid electron transfer from the electrode. Apart from acting as a support, some carbon materials provide intrinsically good adsorption capacity for CO₂, for example, rich oxygen-containing groups present in the graphene oxide (GO) [89], or amine pendant groups present in the nitrated graphene oxide (N-GO) [90]. Porphyrins, corroles, and phthalocyanines are chemisorbed on these carbonaceous materials via non-covalent interactions involving the delocalized π electrons of the macrocyclic ring. To obtain a catalyst with maximum selectivity and current density, the ratio between molecular catalyst and support has to be optimized since too high content of molecular catalyst will lead to its agglomeration via ' π - π ' stacking to form crystals, which are unstable and less efficient for CO₂R. In practice, the adsorption is done by mixing a dispersion of the carbon support with the molecular catalyst in the solution. The loss in the color of the porphyrin or phthalocyanine solution is a clear indication of the successful impregnation of the carbon support with the catalyst [91],

resulting in the catalyst ink which, in combination with a binder such as Nafion, is finally deposited onto, commonly, high surface area carbon paper.

Huang *et al.* immobilized **Co-Pc-1** on Graphene oxide (GO) in a controlled manner by a facile and rapid mixing of **Co-Pc**/sulfuric acid solution with a GO aqueous system (Scheme 5). The optimal nanocomposites with 1:1 ratio content of **Co-Pc**/GO produced CO achieving FE_{CO} of 96% and j_{CO} of 4.6 mA.cm⁻² at -1.0 V vs. RHE. FE_{CO} of 80% was maintained for 2.7 h at -0.9 V reflecting reasonably good stability and durability of the nanocomposite [92]. Gong *et al.* could rise the j_{CO} of heterogenized **Co-Pc** by taking advantage of the CO₂ rapid adsorption in 3D graphene structures [93]. In particular, **Co-Pc-1** was anchored on high surface area nitrogen-doped hollow carbon spheres (N-HCS) with a final content of Co of 0.49 wt% (measured by the ICP method). The **Co-Pc**/N-HCS catalyst produced CO at -0.82 V vs. RHE with j_{CO} of 20.47 mA.cm⁻² (FE_{CO} = 96%) [94]. In another work, immobilization of **Co-Pc-1** with 3D nitrogen-doped vertical graphene arrays (NVG) was grown *in-situ* on carbon cloth (CC) without any binder (Kong *et al.*) material. This approach fetched strong Co-N covalent bonds between N of NVG and Co of **Co-Pc**. The **Co-Pc**/NVG/CC catalyst exhibited FE_{CO} higher than 97.5% over a wide potential range ($\approx$ -0.78 to -1.18 V vs. RHE) potential and an outstanding j_{CO} of 46.1 mA cm⁻² at -1.08 V vs. RHE [95]. Carboxylated graphene oxide (GO-COOH) was used to chemically attach a Co-Pc-bearing tetrasulfonamide group (**Co-Pc-8**) for photoelectrochemical (PEC) CO₂ reduction (Scheme 5). Under sun-light illumination in an aqueous electrolyte, the **Co-Pc**/GO-COOH catalyst



Scheme 5. Metallophthalocyanines used for CO₂ reduction catalysis.

needed -1.0 V vs. Ag/AgCl of external bias to produce formate with high selectivity ($FE_{\text{formate}} = 83.9\%$) at a rate of $2.35 \text{ mmol.cm}^{-2}.\text{h}^{-1}$, with j_{formate} of -1.5 mA.cm^{-2} . Intermolecular interactions between GO-COOH and **Co-Pc-8** originate the necessary high mobility of the photogenerated electrons on the hybrid surface [96]. In a homogeneous medium, **Co-Por-14** most commonly produces CO as a CO_2R product as usual, however, it could be extended to CH_4 by using the same catalyst with a 3D graphene support containing free amine functional groups. **Co-Por-9** catalyst was hosted in an azo-bonded porous covalent graphene framework (CGF) and it produced CH_4 at -0.72 vs. RHE in aqueous media, with an impressing FE_{CH_4} of 21% (Scheme 5). It was envisioned that the **Co-Por-9** catalyst in the supported form promoted the formation and stabilization of CO^* and CHO^* intermediates which later underwent hydrogenation to CH_4 [90]. To enhance the catalytic performance of **Fe-Por-1**, Gotico *et al.* designed the bio-inspired **Fe-Por-11** containing urea functions in the second sphere and explored its CO_2R activity in DMF which as discussed in the previous section (Scheme 1) [55]. In a later work, the same group chose multiwall carbon nanotubes (MWCNTs) to immobilize the catalyst. The **Fe-Por-11**/MWCNT composite displayed excellent selectivity towards CO production with FE_{CO} of 90% at a potential of -0.78 V vs. RHE, with 0.7 mA.cm^{-2} highly stable current for 2 h test, which represents an increase of 30% in FE_{CO} with respect to **Fe-Por-1**/MWCNT composite, proving the impact of added hydrogen bond donor groups in the heterogenized form of the catalyst [91]. In another work, Abdinejad *et al.* enhanced the activity of the iron porphyrin dimer [**FePor**]₂-**2-1** by non-covalent immobilization of the catalyst on also MWCNTs, achieving a FE_{CO} of 89% at -0.88 vs. RHE with a high j_{CO} of 16 mA.cm^{-2} , which was stable over 6 h (Scheme 2). These results implied a 2-fold enhancement of FE_{CO} and a reduction of the overpotential for CO production concerning the homogeneous system [97]. These reports certify that molecular-level architecture design can be further extended to heterogeneous catalysis. Other **Fe-Por**, **Co-Pc**, **Ni-Pc**, and their derivatives have been immobilized in the carbon powder [98,99] [100], graphdiyne/graphene [101], functionalized CNTs [102–106], nanorods and nitrogen-doped GO [107]. For the reduction of CO_2 to C_{2+} products, **Cu-Pc** and derivatives have been used in CNTs [108]. Their performance as heterogenized molecular catalysts working in a flow-cell configuration will be discussed in the next section.

Heteronization of metallocorrole complexes to the carbon materials for efficient CO_2R has also been carried out in recent times. Roy and Schöfberger group reported the formation of CO_2 to alcohols, formaldehyde, carboxylates, and hydrogen using a cobalt(III) corrole complex **Co-Cor-1** modified carbon paper electrode in moderate acidic (pH 6) medium (Scheme 4) [109]. The presence of three *meso* tetrafluorophenyl groups containing polyethylene

glycol residues helped in anchoring and equally distributing the catalyst on the electrode surface. Here also, Co(I) corrole acts as an electron-rich super nucleophile to activate the CO_2 molecule. Electrolysis at -0.8 V vs. RHE showed the formation of ethanol with the highest FE_{ethanol} (48%) in the liquid phase whereas in low overpotential (-0.515 V vs. RHE) methanol was formed in high FE_{ethanol} (59%). The same research group later reported the formation of acetic acid from CO_2 using Mn(III) corrole complex **Mn-Cor-1** immobilized on a carbon paper electrode, having the same polyethylene glycol-based substituents (Scheme 4) [110]. Electroactive Mn(II) corrole binds the CO_2 and reduces it to acetic acid with a selectivity of 63% (TOF = 8.25 h^{-1}) in a moderate acidic aqueous medium (pH 6). The mechanism of acetate formation was proposed via the formation of an oxalate-type intermediate follow by its reduction. Photoelectrochemical reduction of CO_2 with the **Co-Cor-4** anchored onto MWCNT-coated Si/TiO₂ photoelectrode was recently reported (Scheme 4) [111]. Under 100 mW.cm^{-2} light irradiation, Si/TiO₂/CNT/**Co-Cor-4** hybrid photocathode exhibited $250 \mu\text{A.cm}^{-2}$ photocurrent with 60% FE_{CO} in 0.1 M KHCO_3 aqueous solution.

Metal-organic frameworks

Reticular-type porous materials such as Metal-organic frameworks (MOFs) are regarded as excellent materials for CO_2 reduction given their large surface area and high porosity, which facilitates mass and charge transport [112]. In the field of CO_2 utilization, MOFs were initially used for CO_2 capture and storage [113] but have also been later utilized as a catalyst for the CO_2R reaction. Compared to immobilization by adsorption, the MOF approach offers the opportunity to avoid aggregation and deactivation processes in the catalysts mainly due to the covalent attachment of the framework units. Hod *et al.* deposited films of MOF-525 on FTO by electrophoresis and hosted **Fe-Por** sites. **Fe-Por-2**/MOF/FTO in organic electrolyte produced CO at -1.3 V vs. NHE ($FE_{\text{CO}} = 41\%$) with a total j of 5.9 mA.cm^{-2} in presence of 2,2,2-trifluoroethanol as proton source, although the catalyst degraded within 30 minutes (Scheme 1) [114]. Later, Xin *et al.* improved the electron-transfer ability within the MOF by inserting the electron-conductive polypyrrole (PPy) molecule into the porous structure of a MOF-545 hosting **Co-Por-2** through *in-situ* polymerization of pyrrole. The catalyst PPy/**Co-Por-2**/MOF yielded CO as the main product at -0.8 V vs. RHE ($FE_{\text{CO}} = 98\%$) with a high j_{CO} of 9.5 mA.cm^{-2} stable over 10 h in aqueous electrolyte [115]. To use MOFs for the obtention of C_2 products, Zhou *et al.* prepared a **Cu-Por** framework by assembling **Cu-Por-2** building blocks through coordination with Cu ions (Scheme 1). The catalyst produced hydrocarbons, predominantly CH_4 and C_2H_4 , with a total FE of 74% at -1.4 V vs. RHE in an aqueous electrolyte. FE_{CH_4} reached

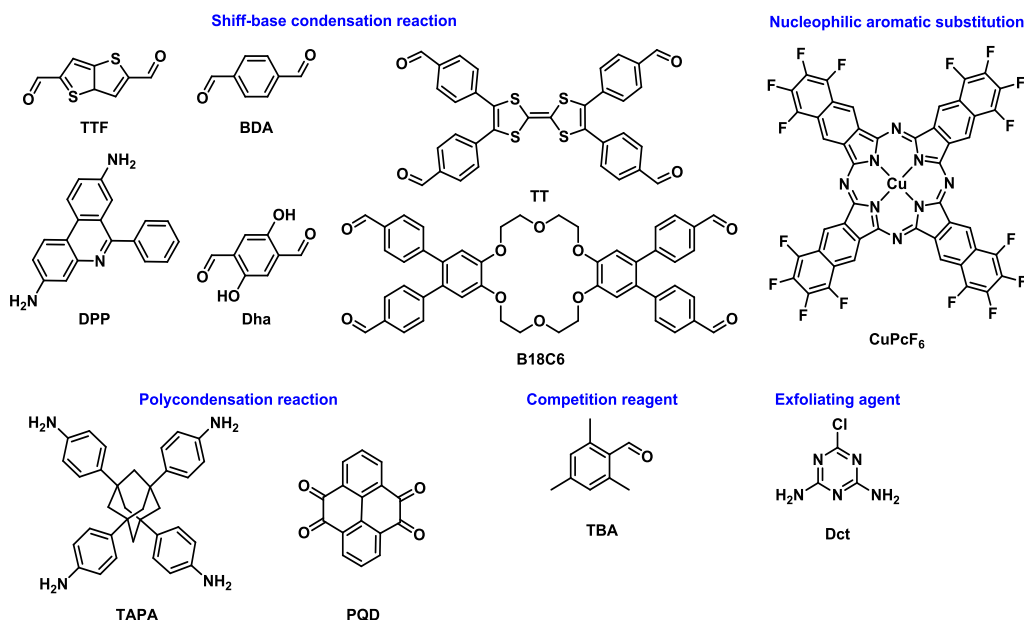
56% at -1.4 V vs. RHE while the highest FE_{C₂H₄} of 32% was achieved at -1.1 V vs. RHE [116].

Photoactive components also inserted in the structure of MOFs have enabled their use in photocatalytic CO₂R. For instance, a photoactive Indium-TCPP MOF hosting **Fe-Por** dispersed in ethyl acetate using L-ascorbyl palmitate as the electron sacrificial agent produced CO at a rate of 144 μmol.g⁻¹.h⁻¹ with 99.5% selectivity after 24 h [117]. In an attempt to incorporate dual catalytic sites into Ni-MOF, Ding *et al.* assembled Nickel(II) terpyridinyl porphyrin **Ni-Por** with the Ni metallation at the peripheral terpyridinyl ligand to yield Ni-bis(terpyridine) complex (Scheme 1). The photocatalytic activity of the **Ni-Por**/NiTPy/MOF was tested with the help of [Ru(bpy)₃]²⁺ as the photosensitizer in acetonitrile/water electrolyte. It produced CO in a 98% selectivity with a high rate of 3984 μmol g⁻¹.h⁻¹ [118]. For photoelectrochemical CO₂R, Wei *et al.* used MOFs to functionalize a Si/TiO₂/Cu photocathode covered by Cu(OH)₂ nanowire arrays that served as the precursor for the *in-situ* growth of Cu-TCPP/MOF from **Cu-Por-2** (Scheme 1). The **Cu-Por**/MOF/TiO₂/Cu/Si photocathode produced CO with a maximal FE_{CO} of 87% at -1.8 V vs. NHE with a current density of 10 mA.cm⁻² in a mixed acetonitrile/water electrolyte under visible light irradiation [119].

Covalent organic frameworks

Covalent organic frameworks (COFs) are well-known for their well-defined 2D or 3D highly porous polymers. Diversely to MOFs, COFs are made from both organic nodes and linkers which are connected through strong covalent bonds [120]. Metal types (e.g., Fe, Co, Cu, and Ni, etc.) can be modified into the structure of COFs

structures to impart them with tunable catalysis centers, such as metalloporphyrins or phthalocyanines. For instance, Zhang *et al.* prepared a hollow tubular CuPor/Dha/COF by Schiff base condensation of 2,5-dihydroxyterephthalaldehyde (Dha) (Scheme 6) and **Cu-Por-8** in the presence of 2,4,6-trimethylbenzaldehyde (TBA) as a competition reagent (Scheme 1 and Scheme 6). Under light irradiation and in the absence of any co-catalyst or hole scavenger in an aqueous electrolyte, **Cu-Por**/Dha/COF produced CO and CH₄ at 15.9 μmol.g⁻¹.h⁻¹ and 13.7 μmol.g⁻¹.h⁻¹ production rates, respectively [121]. Wang *et al.* similarly used 3,8-diamino-6-phenyl-phenanthridine (DPP) (Scheme 6) and **Co-Por-3** as building blocks to construct a **Co-Por**-based COF with an enlarged inter-layer spacing. The **Co-Por**/DPP/COF performed photocatalytic CO₂ to CO conversion in acetonitrile/water/TIPA electrolyte in presence of [Ru(bpy)₃]Cl₂ as the photosensitizer. It achieved a maximum rate of 10200 μmol.g⁻¹.h⁻¹ for syngas production (CO/H₂ selectivity of 82%) upon 2 h visible light irradiation [122]. Cheung *et al.* prepared a thin film of COFs *in-situ* on carbon cloth from **Fe-Por-8** and Dha precursors via a solvent-free synthesis method. **Fe-Por**/Dha/COF exhibited reasonable FE_{CO} near 80% at -2.2 V vs. Ag/AgCl over 3 h in acetonitrile with 0.5 M trifluoroethanol as a proton source [123]. Han *et al.* succeeded in constructing a 3D COF by the polycondensation between **Co-Pc-5** and 1,3,5,7-tetra(4-aminophenyl) adamantane (TAPA) (Scheme 1 and Scheme 6). The unique 3D porous structure of **Co-Pc**/PI/COF ensured an electrochemical surface concentration as high as 32.7% of the total **Co-Pc** subunits acting as active centers. As a result, the catalyst mixed with carbon black achieved an outstanding *j*_{CO} of 31.7 mA.cm⁻² at



Scheme 6. Molecular structure of the building blocks used in the metalloporphyrin and phthalocyanine-based MOF and COF synthesis.

-0.9 V vs. RHE with FE_{CO} of 96% and stability over 20 h [124]. In recent times, this example can be regarded as the best among all the reported COFs for CO_2R measured at similar conditions. Further, the same group employed **Co-Pc-5** to polymerize with 1,4-phenylenediamine (PD). The **Co-Pc-5**/PD/COF catalyst, mixed with carbon black and supported on carbon fiber paper, exhibited a FE_{CO} of 97% at -0.80 V vs. RHE with a significant j_{CO} of 21.2 $\text{mA}\cdot\text{cm}^{-2}$ at -0.9 V vs. RHE in aqueous electrolyte [125]. Wu *et al.* reported two **Co-Por**-based donor-acceptor (D-A) COFs comprising of sulfur-containing aromatic heterocycles as the donor and **Co-Por-8** as the CO_2R catalytic unit (Scheme 1). Tetrathiafulvalene (TTF) [126] and thiophene-2,5-dicarbaldehyde (TT) [127] were used as donor moieties (Scheme 6). Both COFs possessed a 3 to 4 times greater charge carrier mobility than that of reference COF-366-Co, which later accounted for an increase of 20–30% of the FE_{CO} and of 2.5–3 times of the current density in aqueous electrolyte. **Co-Por**/TTF/COF showed the highest FE_{CO} of 95% at -0.7 V vs. RHE and a peak j_{CO} of 6.88 $\text{mA}\cdot\text{cm}^{-2}$ at -0.9 V vs. RHE, whereas **Co-Por**/TT/COF showed higher j_{CO} of 7.28 $\text{mA}\cdot\text{cm}^{-2}$ at a lower potential (-0.7 V vs. RHE) and had longer stability. A hydrophilic COF with efficient electron transfer capability was designed by An *et al.* [128]. They have introduced the crown ether B18C6 in the backbone of a **Co-Por-8**-based COF (Scheme 6). The **Co-Por**/B18C6/COF catalyst displayed enhanced CO_2 binding ability and produced CO with a maximum FE_{CO} of 93.3% at -0.7 V vs. RHE and a highest j_{CO} of 9.45 $\text{mA}\cdot\text{cm}^{-2}$ at -1.0 V vs. RHE. Instead of changing the building blocks to achieve higher conductivity, another **Co-Por**-based COF was developed with highly conductive MWCNTs via a covalent bonding [129]. *In-situ* polymerization of porphyrin **Por-3** with *p*-phenylenediamine in the presence of amino-functionalized MWCNTs, followed by coordination with Co(II), yielded the **Co-Por**/COF/CNT material. In aqueous media, it produced CO (FE_{CO} = 99%) at -0.6 V vs. RHE, stable for 50 h of catalysis, and maximum j_{CO} of 18.77 $\text{mA}\cdot\text{cm}^{-2}$ at -1.0 V vs. RHE. In another strategy, Huang *et al.* explored the robust phenazine linkage to connect **Co-Pc** catalytic sites into a fully π -conjugated material. **Co-Pc**/PDQ/COF was obtained via the polycondensation of **Co-Pc-10** with 4,5,9,10-pyrenediquinone (PDQ) (Scheme 5 and Scheme 6). Later it was mixed with carbon black for electrocatalysis and furnished CO at -0.66 V vs. RHE (FE_{CO} = 96%) with a high j_{CO} of 22 $\text{mA}\cdot\text{cm}^{-2}$ stable over 24 h [130]. Another highly stable linking strategy was reported by Yue *et al.*, who adopted the irreversible dioxin linkage for the construction of a bimetallic phthalocyanine-based COF. The **Co-Pc-6**/CuPcF₆/COF was obtained via nucleophilic aromatic substitution reaction of CuPcF₆ (Scheme 6) with **Co-Pc-6** (Scheme 5). The resulting COF exhibited a FE_{CO} peak value of 97% at -0.62 V vs. RHE with steady j_{CO} around 15.2 $\text{mA}\cdot\text{cm}^{-2}$ over 1 h of electrolysis in an aqueous media [131].

Covalent organic polymers

Covalent Organic Polymer (COPs) are known for their better stability and a higher tolerance for monomers compared to the MOFs/COFs. In addition, COPs contain linkers that promote the delocalization of π -orbitals within the whole structure, which turns them into very promising competitors in the field of photocatalysis and electrocatalysis as an alternative to MOFs/COFs. He *et al.* prepared COPs with **Fe-Por** sites in two steps; condensation of 4,4'-biphenyldicarboxaldehyde with pyrrole followed by partial complexation of the porphyrin with Fe^{3+} ions to afford Por/**Fe-Por**/COP. After 5 h of visible light irradiation in DMF electrolyte with the presence of $[\text{Ru}(\text{bpy})_3]^{2+}$ as photosensitizer and TEOA as hole scavenger, Por/**Fe-Por**/COP furnished 28,654 $\mu\text{mol}\cdot\text{g}^{-1}$ of syngas ($\text{CO}:\text{H}_2$ with a proportion of 1:2) mixture suitable for the Fischer-Tropsch synthesis [132]. Yu *et al.* used an acetylene bridging unit 4,4'-dibromobiphenyl (DBBP) to assemble **Co-Por-12** through the Sonogashira cross-coupling reaction (Scheme 1). Upon visible-light irradiation in acetonitrile/water solution, the **Co-Por**/DBBP/COP generated CO with a rate of 286.7 $\mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$ (90.4% selectivity) for 25 h without any photosensitizer [133]. Alternatively, Wang *et al.* chose a N'NN'-pincer-ligand (2,6-bis(5-amino-1H-benzimidazol-2-yl)pyridine, N_3 polymer) to obtain a novel **Co-Por-3**/ N_3 /COP which was used for CO_2R after supporting on CNTs. It produced maximum CO (FE_{CO} = 96%) at -0.50 V vs. RHE for 24 h maintaining a current density of 15 $\text{mA}\cdot\text{cm}^{-2}$. The synergistic effect for the favorable electron transfer between the electron-rich N_3 ligands to the Co centers of the **Co-Por** resulted in a better activity [134]. To attain better polymerization between the porphyrin and CNT unit, Y. Wang *et al.* used the Hay coupling to polymerize **Co-Por-19** on CNTs (Scheme 1). The resulting **Co-Por**/COP/CNTs produced CO with a high FE_{CO} of > 95% in a wide potential range of -0.57 to -0.87 V vs. RHE over a long period (~100 h electrolysis) [135]. In another work, Wang *et al.* employed the Sonogashira coupling reaction, *in-situ* on CNTs, between bromo-substituted **Co-Por-10** and ethynyl-substituted **Co-Por-28** to yield **Co-Por**/COP/CNTs (Scheme 1). The catalyst mixed with carbon black produced CO at -0.55 V vs. RHE (FE_{CO} = 94%) along with a j_{CO} of 9 $\text{mA}\cdot\text{cm}^{-2}$ for 24 h [136].

Cobalt phthalocyanine (Co-Pc) embedded within pyridine polymers

Encapsulation of catalysts into coordinating polymers is an effective immobilization strategy for having efficient CO_2R . McCrary's group has made substantial progress with this approach by focusing on the activity of **Co-Pc**/P4VP polymer-catalyst composite systems comprised of poly-4-vinylpyridine (P4VP) polymers and **Co-Pc** [137]. **Co-Pc-1** was encapsulated within the poly-4-vinylpyridine (P4VP) or poly-2-vinylpyridine (P2VP) polymers by simply solution phase mixing and

subsequent deposition on edge-plane graphite disc electrodes (Scheme 5). The protonation of the pyridine residues from the secondary and outer coordination spheres of the Co center could increase the availability of protons around the catalyst active sites. Moreover, the synergistic relationship between axial coordination and the chemical environment imposed by the P4VP membrane led to a dramatic enhancement in j_{CO} value to 2.0 mA.cm⁻², compared to CoPc/P2VP (0.39 mA.cm⁻²) [138]. Later, mechanistic analysis with **Co-Pc-1**/P4VP suggested that axial-coordination of pyridyl/pyridine to **Co-Pc-1** changes the rate-determining step of the reaction and proton delivery to the Co-Pc active sites is controlled by a polymer-based proton relay mechanism involving the pyridyl moieties [139]. Recently, the same group has systematically explored the incorporation of graphite powder (GP) to enhance the inefficient charge transport capacity of the **Co-Pc**/P4VP system. **Co-Pc-1**/P4VP/GP catalysts with different ratios of **Co-Pc-1**:P4VP:GP were prepared by sonication followed by a centrifugation step that concentrated the CoPc/P4VP system near the GP facilitating physisorption to the particles. **Co-Pc-1**/P4VP/GP produced CO with a current density of $j_{CO} \approx 6$ mA.cm⁻², three times higher than that of analogous **Co-Pc-1**/P4VP systems. However, in this case, lowering in FE_{CO} to 83% was also noted [140].

Functionalization on semiconductor materials and metal nanoparticles

Metal oxide semiconductors: Metal oxides have an affinity toward various functional groups, such as phosphonic acid and a carboxylic acid, which can be exploited for catalyst anchoring. In this regard, various metal oxide has been employed as a top cathode layer, although titanium dioxide (TiO₂) is the most common since it is cheap and abundant. Under visible light irradiation in water, Wu *et al.* used thin TiO₂ film modified with **Fe-Por-1** for CO₂R (Scheme 1). It produced CO and CH₄ without using any sacrificial electron donor. The modification with **Fe-Por-1** increased CH₄ formation up to 20.72 μmol.g⁻¹.h⁻¹ with 56.1% selectivity over CO and stability up to 60 h. The authors proposed that the synergy between the oxygen vacancies (OVs) of TiO₂ and **Fe-Por-1** promoted electron-hole separation and enhanced CO adsorption ensuring the further reduction of CO to CH₄ [141]. Ferreira de Brito *et al.* reported methanol and ethanol in PEC CO₂R formation (0.35 mM and 0.033 mM, respectively) for the first time using a Ti/TiO₂ nanotubes electrode as a support after immobilizing **Cu-Por-18** by simple impregnation method (Scheme 1). PEC was carried out under -0.8 V vs. Ag/AgCl and UV-vis light illumination in an aqueous electrolyte for 2 h. A mechanism involving photoactivation of the TiO₂ support and electron transfer to the Cu-porphyrin was proposed [142]. Recently, Reisner and co-workers immobilized **Co-Pc-11** bearing four phosphonic acid anchoring

groups for a covalent stable link on mesoporous TiO₂ (Scheme 5). The Ti/*meso*-TiO₂/**Co-Pc-11** assembly produced CO with a maximum FE_{CO} of 91% over 2 hours of electrolysis at -1.09 V vs. SHE with j_{CO} of 1.3 mA.cm⁻² in an aqueous electrolyte. Furthermore, this hybrid assembly was utilized in solar-driven CO₂ conversion by integrating with p-type silicon (Si) photoelectrode. Si/*meso*-TiO₂/**Co-Pc-11** was tested under PEC conditions (visible light illumination and -0.53 V vs. SHE applied potential) and produced CO (21 μmol.cm⁻²) and 11 μmol of H₂ over the course of 12 h with a total FE_{CO} of 56%, maintaining a stable photocurrent of approximately 150 μA.cm⁻² [143]. Ma *et al.* designed a hyper-crosslinking porphyrin-based polymer grown *in-situ* on hollow TiO₂ to test its activity for O₂-tolerant photocatalytic CO₂ reduction. After coordination with palladium (II), the hybrid system **Pd-Por-1**/hs-TiO₂ showed the production of 24.3 μmol.g⁻¹ of CH₄ after 2 h of UV-Visible light irradiation in water with 5 vol% O₂. The enhanced activity is attributed to high CO₂ adsorption and reduction over O₂ provided by the π -conjugated structure of the porphyrin polymer [144]. Mennel *et al.* reported two covalently attached hybrid assemblies on FTO substrate by reacting alkyne-modified porphyrins **Fe-Por-36** and **Co-Por-36** with azide-terminated phosphate SAMs (Scheme 1). In aqueous electrolytes at high overpotentials, **Co-Por-36**/SAM/FTO favored CO production and **Fe-Por-36**/SAM/FTO yielded HCOOH production by virtue of fast electron transfer [145].

Semiconductor nanoparticles: Quantum Dots (QD) are excellent photosensitizers and behave as excellent sources of transferring photoelectrons to the porphyrins [146], especially for porphyrin-functionalized QDs assemblies in the water [147]. Taking advantage of these properties, Arcudi *et al.* designed a photocatalytic system of heavy metal-free CuInS₂/ZnS colloidal QDs as photosensitizers for **Co-Por-9** (Scheme 1). Under blue-light illumination, **Co-Por-9**/CuInS₂/ZnS hybrid catalyst showed an unprecedented performance for CO production (100% selectivity) with TON > 80000 in water in presence of tris(carboxyethyl)phosphine and Na-ascorbate as the sacrificial reductants [148]. For instance, this is a remarkable performance obtained with **Co-Por-9** compared to the systems reported by the Sakai group as described earlier [75].

Copper nanoparticles: Cu-based heterogeneous materials are distinctive in the CO₂R catalysts family due to their unique ability to reduce CO₂ to C₂₊ products. However, optimizing the activity towards one particular product formation remains a crucial challenge since multiple intermediates are involved in the CO₂ to C₂₊ conversion pathway. In general, CO formed on Cu active sites has been identified as a crucial intermediate that favors the necessary C-C coupling for the C₂₊ products formation [149–151]. With this idea, Sargent and co-workers reported the first **Fe-Por-1**-Cu composite that was able to steer the selectivity towards ethanol

when tested in a flow-cell configuration (Scheme 1) [37], which is described in the next section. Later, Yan *et al.* used Cu-MOFs to encapsulate **Fe-Por-2**, which were later converted into Fe-Por-functionalized Cu nanocorals by pre-reduction of the Fe-TCPP/Cu-MOF. The generated Fe-TCPP/Cu hybrids containing both Cu and Cu₂O in combination with Fe-Por sites were able to improve 2-fold C₂H₄ production concerning the non-functionalized Cu nanocorals, demonstrating a FE_{C₂H₄} of 33.42% at -1.17 V vs. RHE with *j*_{C₂H₄} of 4.5 mA.cm⁻² [152]. Recently, ethylene production was substantially increased by another hybrid catalyst designed by Buonsanti and co-workers consisting of Cu nanocubes (NCs) and different Fe porphyrins. The Fe porphyrin with the highest π -delocalization, an iron(III)-chloridotetraphenyl porphyrin (**Fe-TBTPP**), produced CO with the highest FE_{CO} of 91% at -0.55 V vs. RHE supported on CNTs (Scheme 2). Although the addition of all the Fe porphyrins to Cu NCs caused an enhancement of C₂H₄ production concerning bare Cu NCs, the hybrid **Fe-TBTPP**/Cu reached the highest FE_{C₂H₄} of 36% at -1.05 V vs. RHE with *j*_{C₂H₄} of 1.3 mA.cm⁻², increasing the FE_{C₂H₄} of Cu NC by almost the double. No electronic nor chemical change of the Cu surface was observed in the hybrid system, proving that the overall selectivity and activity is purely governed by the tandem catalytic process [153].

Carbon nitride: As an alternative to metal-oxides, Carbon nitride (C₃N₄) emerges as a metal-free visible-light responsive material with no toxicity, low cost, appropriate band positions, and high (photo)chemical stability. Sun *et al.* constructed ultrathin transitional **Cu-Pc-1**/CN heterojunctions, via a controlled surface hydroxyl-induced assembly process (Scheme 5). However, they could only achieve a very low production rate for CO and CH₄ of 1.5 μ mol.g⁻¹.h⁻¹ and 0.4 μ mol.g⁻¹.h⁻¹, respectively, under visible light irradiation in an aqueous medium [154]. In another attempt, Li *et al.* prepared a mesoporous graphitic-C₃N₄ (mpg-C₃N₄) with surface-bound **Co-Pc-1** 0D nanocrystals(nc) by *in-situ* polymerization of 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) on mpg-C₃N₄. Under UV-visible light illumination in acetonitrile with a sacrificial electron donor, the nc**CoPc-1**/mpg-C₃N₄ catalyst produced a good amount of CO (6052 mmol.g⁻¹) in 8 h [155]. Further, the same group reported the use of gC₃N₄ nanosheets/CNTs as support for ncCoPc and achieved CO₂ to CO conversion (FE_{CO} = 95%) with *j*_{CO} of 22 mA.cm⁻² at 0.8 V vs. RHE in aqueous media [156]. In another attempt, Liu *et al.* doped gC₃N₄ with phosphorous to improve light harvesting property and used it as a support to immobilize **Co-Pc-1** via π - π stacking interactions. Under visible light irradiation in water, the **Co-Pc-1**/P-gC₃N₄ catalyst produced 295 μ mol.g⁻¹ of CO in 24 h, with ~ 90% selectivity over H₂ [157].

Metalloporphyrinoids in CO₂ reduction electrolyzer and photochemical reactors

Industry scale production of CO₂R products needs highly efficient and durable (> 24 h) catalysts operating at high current densities (> 100 mA.cm⁻²) [158]. Typical H-type electrochemical cells suffer from mass transport limitations due to the inherently low diffusion and solubility of CO₂ in aqueous media. For such purpose, continuous flow reactors emerge as an alternative that offers a continuous circulation of the CO₂-saturated aqueous electrolyte [159] to the cathode at regular intervals. However, it has to be taken into account that higher CO₂ concentration at the electrocatalyst interface and other aspects result in a change in both thermodynamic and kinetics aspects of the CO₂R [5,160,161]. The most common CO₂ flow cells have OER at the anode, which is separated from the cathode by a polymer electrolyte membrane (PEM) that facilitates ion flow while preventing product crossover [162]. The catalyst is supported on highly-porous gas diffusion layers (GDLs) situated between the electrode and the membrane. A typical GDL consists of a porous carbon-containing matrix that allows gas transport but limits the transport of liquids [163]. However, for CO₂R, the GDLs often suffer from extremely poor stability [163]. In a nutshell, the CO₂R electrolyzer performances should be optimized with both the cathodic and anodic compartments reactions in a combined way such that, under the same conditions of electrolyte, pH and current density, the overall cell potential is decreased without scarifying the performances.

Earlier CO₂R electrolyzers were designed to produce CO typically relying on heterogeneous metal catalysts, such as gold, silver, or copper. Of late, the high activity and selectivity of metal phthalocyanines and porphyrins for CO₂ to CO conversion has motivated their use in electrolyzers too. The use of metalloporphyrinoids deposited on a GDL was evaluated dating back to the late 1980s and 1990s when partial currents for CO production reached up to 55 mA.cm⁻² for **Fe-Por-1** in aqueous electrolyte (Scheme 1) [164–169]. However, the progress in heterogenization strategies, as well as the improvement in the design of flow cells, have attained > 90% selectivity at high current densities of >150 mA.cm⁻² [39,170] in the past years. In this section, we will review the recent studies carried on with metal porphyrins and phthalocyanines in flow-cell configuration, paying special attention to the immobilization strategy on the GDL and to the working conditions.

As discussed previously, *immobilizing molecular catalysts on carbon supports* is an effective way to enhance conductivity and obtain higher current densities during CO₂R in aqueous media. This strategy has been extended to flow-cell configurations, in which the molecular catalyst is combined with carbon support material and Nafion binder, *i.e.*, the catalyst ink is deposited on the GDL by different techniques such as

spray-coating or drop-casting. Two types of carbon supports have been mainly used over the last years, namely carbon powder and CNTs. Berlinguette and co-workers used spray-coated **Co-Pc-1** based ink on GDL to evaluate their performance in a zero-gap membrane reactor (Scheme 5). The catalysts produced CO with FE_{CO} of > 95% maintained at current densities of 150 mA.cm⁻² with an overall two-electrode cell voltage (E_{cell}) of 2.5 V. The incorporation of phenol in the catalyst ink helped to increase the current density to 200 mA.cm⁻² while the FE_{CO} could be held at >90% over 8 hours of electrolysis at 50 mA.cm⁻² [39]. In further work in collaboration with Robert's group, a new **Co-Pc-3** bearing a trimethyl ammonium moiety and tert-butyl groups was reported as highly effective CO₂R electrocatalysts in a wide range of pH conditions (Scheme 5). In a commercial flow-cell electrolyzer in alkaline KOH electrolyte (pH 14), **Co-Pc-3** supported on a GDL electrode reached a peak j_{CO} of 165 mA.cm⁻² at -0.92 V vs. RHE with FE_{CO} of 94% and could sustain a j_{CO} of 70.5 mA.cm⁻² during 10 h at an applied potential of -0.65 V vs. RHE [98]. Later post-mortem analysis of the **Co-Pc** based systems in a flow-cell configuration by Wan *et al.* revealed that loss in FE_{CO} originates from demetallation of the CoPc, leaching due to weakening of the interaction with the carbon black support, and morphological changes in the GDL micropores [99]. Wu *et al.* dispersed **Co-Pc-1** on CNTs paying extra attention to the supporting methodology to avoid **Co-Pc** aggregates and obtain molecularly dispersed catalyst sites. The optimal **Co-Pc-1**/CNT catalyst was mixed with carbon black and Nafion and air-brushed onto a carbon paper GDL. In alkaline 1 M KOH, the j_{CO} could reach up to 200 mA.cm⁻² with a considerable FE_{CO} of 93% [104]. Su *et al.* immobilized a tetraamino-functionalized **Co-Pc-9** on CNTs by diazonium reaction and then transformed the amino groups into quaternary ammonium groups **Co-Pc-2** which is proved to impart better electron-withdrawing effect. In alkaline 1 M KOH, the **Co-Pc-2** catalyst showed an outstanding j_{CO} of 239 mA.cm⁻² at -0.7 V vs. RHE with a great CO selectivity (FE_{CO} = 95.6%) [103]. In an alternative approach, the molecular phthalocyanine **Co-Pc-1** was immobilized with amino, hydroxyl, and carboxyl functionalized CNT via non-covalent interactions like Co-N/Co-O axial coordination [102]. In these cases, also the catalyst ink was prepared by simply mixing **Co-Pc-1** and functionalized CNTs, and further drop-casted onto a GDL before use. The **Co-Pc-1**/CNT-NH₂ showed the better performance reported so far with a remarkable j_{CO} of 225 mA.cm⁻² with nearly 100% FE_{CO} at -0.6 V vs. RHE, also the system maintained its performance for < 100 h. The j_{CO} could be further increased to 351 mA.cm⁻² by applying -0.7 V vs. RHE with only a 4% loss in CO selectivity. By employing another innovative strategy, Gu *et al.* combined graphene (G) with graphdiyne (GDY), a 2D carbon allotrope with sp- and sp²-hybridized carbon atoms with diacetylene linkages that provide strong electronic

coupling with **Co-Pc**. The **Co-Pc-1**/GDY/G achieved a j_{CO} of 100 mA.cm⁻² with a FE_{CO} of 97% at a modest bias of -0.82 V vs. RHE [101]. Lastly, Li *et al.* attempted to avoid the problematics related to catalyst aggregation by self-assembly of Co-Pc into nanorods with highly ordered molecular alignment. Non-peripheral octamethylsubstituted **Co-Pc-4** nanorods were immobilized on nitrogen-doped reduced GO (NRGO) and deposited on a GDL. In 1 M KOH, the composite exhibited the highest FE_{CO} of 94.1% at -0.6 V vs. RHE, while the j_{CO} reached a modest 56.4 mA.cm⁻² [107].

Chen *et al.* used a **Ni-Pc** with amino groups catalyst (**Ni-Pc-3**) supported on CNTs, producing an ultrahigh current density of 500 mA.cm⁻² at -0.75 V vs. RHE with RHE FE_{CO} of 94.6% in 1 M KHCO₃. Moreover, the catalyst could maintain a current density of 150 mA.cm⁻² for 10 h with CO selectivity over 99.8%. Amino functionalization improved by almost 5 times the current density with respect to NiPc/CNTs under the same conditions [106]. Robert and co-workers used **Fe-Por-16** substituted with trimethyl ammonium groups as a catalyst in a similar flow cell assembly. The best performance was observed with a j_{CO} of 155 mA.cm⁻² at -0.59 V vs. RHE (98.1% CO selectivity) in 1 M KOH as an electrolyte [100]. In another similar study, amino-substituted **Fe-Por-35** exhibited a maximum FE_{CO} of 85% and j_{CO} of 103.8 mA.cm⁻² at -0.8 V vs. RHE in 1 M KOH electrolyte [105].

Although the use of concentrated hydroxide electrolytes (for example 1 M KOH, pH 14) promotes CO₂R by suppressing the competing HER, it converts up to 70% of reacted CO₂ into unreactive (bi)carbonates which decreases both the energy efficiency and the durability of the reactor [171,172]. As an alternative, using CO as a reagent can avoid the formation of (bi)carbonates while increasing the probability of C-C coupling. For instance, Ren *et al.* reported the production of C₂ products from CO with **Cu-Pc-1** immobilized on carbon black in 2 M KOH as the electrolyte using a 'flow-through' configuration with high efficiency. Acetate (FE_{AcO-} = 36%) and C₂H₄ (FE_{C₂H₄} ~ 20%) were found to be the predominant liquid and gaseous product, respectively, at a current density of 200 mA.cm⁻² (j_{C_2+} = 147 mA.cm⁻²) [108].

Integration of metalloporphyrins and metallophthalocyanines into covalent organic frameworks (COF) opens up a suited approach for having extended, ultrathin nanoscale materials with a high number of exposed active sites as previously discussed, and thereby capable of achieving a fairly high current density in flow cell reactor. Rong *et al.* prepared a **Cu-Por-8**/Dha COF and functionalized it with 2,4-diamino-6-cholo-1,3,5-triazine (Dct) (Scheme 6). Amino and triazine groups of Dct worked as an exfoliation agent that induced the exfoliation into nanosheets, besides facilitating the generation of CO around the **Cu-Por** active sites and improving the selectivity for CH₄. Specifically, **Cu-Por-8**/Dha achieved a remarkable FE_{CH₄} of 80% with j_{CH_4} of 220 mA.cm⁻² at

-0.90 V vs. RHE in 1 M KOH electrolyte which is 100 mA.cm⁻² more than the same COF without Dct functionalities (Scheme 6) [173]. Song *et al.* synthesized **Co-Por-8**/BDA/COF via imine condensation and methylated the reduced imine bond to obtain cationic quaternary ammonium groups that weakened the interlayer stacking interaction and helped the exfoliation into nanosheets (Scheme 1 and Scheme 6). The **Co-Por-8**/BDA/COF nanosheets achieved the highest j_{CO} of 300.2 mA.cm⁻² at -0.75 V vs. RHE with 90% FE_{CO} in 1 M KOH electrolyte, while j_{CO} of $\approx$ 200 mA.cm⁻² was stable for 11 h with FE_{CO} above 95%. The catalyst was also applied in a photovoltaic-electrochemical system with Fe/Ni foam as the anode and two commercial GaInP/GaInAs/Ge triple-junction solar cells, giving a solar-to-CO conversion efficiency of $\approx$ 14% [174]. In another work, Wang *et al.* verified the feasibility of hydrogen-bonded organic frameworks (HOFs) as an efficient and stable CO₂R platform by using **Cu-Por-6** decorated with diaminotriazine moiety as the building block. The diaminotriazine moiety acted as a hydrogen bonding network and also as a proton pump to the Cu center. **Cu-Por-6**/HOF exfoliated into large-scale nanosheets and tested in PBS (pH 7) electrolyte. At -1.3 V vs. RHE the catalyst produced mainly CO, however, at -1.6 V vs. RHE, methane production was noted with j_{CH_4} of 183 mA.cm⁻² for 5 h (FE_{CH₄} = 70%) [175]. Li *et al.* adopted bimetallic approaches by using a polymeric **Co-Pc-1**/**Zn-Pc-1** framework with unusual AB layer stacking. The optimal (**Co-Pc-1**)₃/(**Zn-Pc-1**)₁ framework achieved a j_{CO} of 212 mA.cm⁻², almost double as the analogous **Co-Pc** framework, at -0.9 V vs. RHE with FE_{CO} of 90% in 1 M KOH electrolyte. Furthermore, a stable j_{CO} of 200 mA.cm⁻² was maintained at -1.1 V vs. RHE for 9 h [176]. Another alternative approach was adopted by Yasri *et al.*, who assembled metal-protoporphyrin IX (**M-PPorIX**) by direct electropolymerization onto the carbon GDL (Scheme 1). In slightly alkaline conditions (1 M KHCO₃, pH 8.2), **Cu-PPorIX** polymer produced methane and ethylene with FE_{CH₄} of 39.4% and FE_{C₂H₄} of 7.96% at a quite high potential of -2.82 V vs. RHE. On the other hand, **Co-PPorIX** polymer produced only CO with FE_{CO} of 94.2% at -1.32 V vs. RHE with a j_{CO} of about 33 mA.cm⁻² [177]. Sargent and co-workers reported the use of heterogeneous Cu surface as support for **Fe-Por-1** [37] as mentioned in section 4.7. **Fe-Por-1** was immobilized on a Cu electrode sputtered on a hydrophobic porous polytetrafluoroethylene (PTFE) substrate that worked as the GDL. **Fe-Por-1**/Cu composite produced ethanol with a FE_{ethanol} of 41% and $j_{ethanol}$ of >100 mA.cm⁻² at -0.82 V vs. RHE.

Photoelectrochemical cells

In photocatalytic (PC) CO₂R, the efficiency is still far from satisfactory [178] and requires other species besides the catalyst such as photosensitizers or sacrificial electron donors. On the other hand, electrocatalytic (EC) systems

directly convert electrical energy into CO₂R products more effectively and actively, but they require a large external energy supply. Photoelectrocatalysis (PEC) synergistically takes advantage of both PC and EC systems, making simultaneous and full use of external bias and solar energy in an integrated reaction system. Over the last 3 years only two groups have reported a complete tandem PEC system without external bias supply. Reisner's group designed the first tandem PEC device that could perform the pure solar-to-CO conversion. The device consisted of a **Co-Pc**-functionalized perovskite-based photocathode and a BiVO₄ photoanode and achieved a modest efficiency of 0.06% and 0.02% [179]. Inspired by this work, Zhang *et al.* fabricated a perovskite photocathode encapsulated in a carbon plate that protected the perovskite while providing it with a surface with CO₂ binding sites. Then, **Co-Pc-1** supported on CNTs was drop-casted on the surface of the carbon plate and the photocathode was integrated with amorphous silicon (*a*-Si) photoanode for OER. The **Co-Pc-1**/CNT-Perovskite-*a*-Si tandem cell realized unassisted PEC CO₂RR under 1 Sun irradiation with a photocurrent density of $\sim$ 3 mA.cm⁻² and a solar-to-CO energy conversion efficiency of 3.34%, stable for more than 1.5 h [180].

Other systems that combine photovoltaics (PV) with EC and PEC present better efficiencies than the PEC cells described above. For instance, Dong *et al.* designed an artificial photosynthesis cell based on a PV-PEC system that included a BiVO₄ photoanode and a TiO₂ photocathode biased with an external Si solar cell. The TiO₂ was functionalized by dip coating with both aminopropyltriethoxysilane, for enhanced CO₂ capture, and **Ni-Por-17** as the selective catalyst for CO production (Scheme 1). In an aqueous electrolyte, the system produced CH₃OH at a rate of 55.5 μ M.h⁻¹.cm⁻², with E_{cell} of 0.6 V, which is the membrane voltage in plants [181]. In another approach, Kobayashi *et al.* chose a TiO₂ photoanode as the counter electrode of a GDL photocathode, modified with **Ni-Pc-1** (Scheme 5). The TiO₂/**Ni-Pc-1** cell produced CO with a high FE_{CO} of 98% at a very low E_{cell} of 0.8 V under a UV illumination [182]. In another example, Wang *et al.* chose a Co₄O₄ cubane complex-modified BiVO₄ as the ORR photoanode and combined it with a carbon cloth (cc) GDL modified by dip coating with **Co-Pc-7**. Under sunlight illumination in mild alkaline aqueous media (pH 8.15), the BiVO₄-cCo/**Co-Pc7** cell produced CO with FE_{CO} of 87% by applying also 0.8 V (E_{cell}). The applied bias-to-photocurrent efficiency (ABPE) at this E_{cell} was 0.52%, while that of the TiO₂/**Ni-Pc-1** cell was 0.18% [183]. Wang *et al.* constructed a PEC system with a CO₂R photocathode and OER photoanode. In that cell, **Co-Por-3**/N₃ polymer/COP catalyst (already described in previously) is supported on CNTs on a carbon cloth cathode coupled with a NiCo-LDH/Cu-Pc/BiVO₄ photoanode. Upon UV-Vis illumination of **Co-Por-3**/N₃polymer||NiCo-LDH/Cu-Pc/BiVO₄ cell exhibited an enhanced current density of 4.2 mA.cm⁻²

at 1.4 V for CO formation with a relatively low FE_{CO} of 60% [134].

CONCLUSION

This review has looked over the recent advances in the application of metalloporphyrinoids for CO₂R from their use as homogeneous molecular catalysts to large-scale devices passing through diverse heterogenization strategies. The recent studies summarized here have proved that the molecular design (*e.g.*, positively charged substituents, H-bonding donor groups, etc.) enhance the activity and selectivity of metalloporphyrinoids towards CO₂R in the homogeneous phase and can be expanded to the heterogeneous medium and finally in flow-cell-based electrolyzers (Table 1). However, the choice of catalyst, the immobilization strategy, the electrolyte, and the current density, affect the system's overall CO₂R performance feats. For instance, for industry-feasible aqueous medium CO₂R, such parameters need to be optimized to preserve selectivity at acceptable current densities while the durability is not compromised. Considering this diverseness in performance, inspection into the catalytic mechanism, such as the impact of solvation and electrolyte, local electric fields, change in catalyst morphology, and degradation pathways in the heterogenized environment become utmost necessary. For the hybrid materials, the maximum number of reports involve iron porphyrin

and cobalt phthalocyanine systems, although metallocorroles and metallochlorins also show potential for being suitable CO₂R catalysts. Combining metalloporphyrinoids with Cu metals appeared as a lucrative way to generate CO₂ to C₂₊ products via a tandem approach. So far, the best approach is the **Co-Pc** functionalized with quaternary ammonium groups covalently immobilized on CNTs, in terms of both current density and durability indicators.

The current CO₂R chemistry with metalloporphyrinoids still has many challenges to overcome. Large-scale applications of metalloporphyrinoids containing materials such as MOFs and COFs for efficient photo or electrocatalytic CO₂R in water have limitations due to several factors. The low conductivity and strong hydrophobicity of MOFs and COFs at the electrode surface hinder the multiple electrons and proton-assisted CO₂R processes. Moreover, the linker (*e.g.*, carboxylic) stability of MOFs in aqueous alkaline electrolytes (pH 7.2–14) under applied bias becomes another issue for their long-term durability. Recently, imine-based COFs have shown enhanced electronic properties; however, COFs still contain closely packed two-dimensional layer structures of metalloporphyrinoids with strong π - π interactions that inevitably lead to insufficient utilization of the active sites. The use of metalloporphyrinoid catalysts supported on carbonaceous materials in flow-cell CO₂ electrolyzers is limited to long-term stability and achieving high industrial-scale

Table 1. Summary of some best-performing metalloporphyrinoid-based CO₂R catalysts explored under different conditions.

Catalyst/ Precursor	Conditions	CO ₂ R products and performances	Ref.
Fe-Por-24	Acetonitrile/TBAPF ₆ and 1.5 M trifluoroethanol, homogeneous medium	CO, FE = 100%, TOF = 1.3×10^9 s ⁻¹	[52]
Zn-bChl	DMF, 5M H ₂ O, homogeneous medium	CO, FE = 92%, j_{CO} = 2.3 mA.cm ⁻² , 4 h	[28]
Mn-Cor-1	Supported on Carbon paper, CO ₂ saturated 0.1 M phosphate buffer (pH 6)	CH ₃ COOH, FE = 63%, j_{CO} = 0.8 mA.cm ⁻² , TOF = 8.25 h ⁻¹	[110]
Fe-TBTPP	Combined with Cu NPs and supported on CNTs	C ₂ H ₄ Fe = 36%	[153]
Fe-Por-2	Supported on Cu NCs generated from Cu-MOF	C ₂ H ₄ , FE = 33%	[152]
Co-Pc-1	Supported on 3D NVG grown on carbon cloth	CO, FE > 97.5%, j_{CO} = 46.1 mA.cm ⁻²	[95]
Co-Por-14	Supported on Covalent graphene framework	CO, FE = 60%; CH ₄ , FE = 21%	[90]
Co-Pc-8	Covalently bound to carboxylated graphene oxide, photochemical condition	HCOO ⁻ , FE = 83.9%	[96]
Co-Pc-5	Embedded in TAPA-COF mixed with Carbon black	CO, FE = 96%, j_{CO} = 31.7 mA.cm ⁻²	[124]
Co-Por-9	Supported on CuInS ₂ /ZnS colloidal QDs	CO, FE ~ 100%, TON > 80000	[148]
Co-Pc-1	Immobilized on NH ₂ -CNTs, in <i>Flow cell configuration</i>	CO, FE ~ 100%, j_{CO} = 225 mA.cm ⁻² for < 100 h	[102]
Ni-Pc-3	Supported on CNTs, in <i>Flow cell configuration</i>	CO, FE = 94.6%, j_{CO} = 500 mA.cm ⁻²	[106]
Cu-Pc-1	Immobilized on carbon black, in <i>Flow cell configuration with CO feeding</i>	CH ₃ COO ⁻ , FE = 36%; C ₂ H ₄ , FE = ~ 20%; j_{C2+} = 147 mA.cm ⁻²	[108]
Cu-Por-8	Supported on COF nanosheets functionalized with Dct, in <i>Flow cell configuration</i>	CH ₄ , FE = 80%, j_{CH4} = 220.0 mA.cm ⁻²	[173]

current density. More careful inspection of Gas diffusion electrodes (GDE) coated with catalysts becomes essential to address these issues. Improving catalyst implantation on the GDE and modifying its configurations should be looked after with great interest. Finally, for photochemical CO₂R, developing new metalloporphyrins in combination with photosensitizers (preferably 3d metals) and the sacrificial donor is still an ongoing task. In PEC systems, photocathode heteroconjugation between the light-harvesting material and the metalloporphyrinoid catalysts should have synergy in terms of efficient charge separation, wide-range light harvesting, and directed electron transfers to the metalloporphyrinoid systems. This area also requires special attention to study interfacial charge transfer kinetics, photocathode fabrication, and CO₂R performance validation. Ultimately, if the photocathode can be combined with a water oxidation photoanode in a PEC configuration, ideally, there is no need for an externally applied bias. However, there are very few reports for such unbiased CO₂R systems. Moreover, the efficiencies of these devices are still relatively low due to the high onset potential for cathodic and anodic reactions and charge recombination inside the device. All these device parameters need to be optimized with intriguing metalloporphyrinoid design and lower CAPEX to have efficient and durable CO₂R systems in reality.

To further advance in the rapidly evolving CO₂R research field, one should consider the beneficial properties of the 3d transition metal-based porphyrinoids. Still, their large-scale applications in CO₂R reactors have a long way to go. We expect that this review will provide general ideas for the rational designing of efficient metalloporphyrinoid catalysts and help the researcher to root out the existing problem of this emerging field.

Author declaration

The authors declare no competing financial interest.

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